



**HKU
Med**

Clinical Pharmacology of Anti-Platelets and Anticoagulation Therapies

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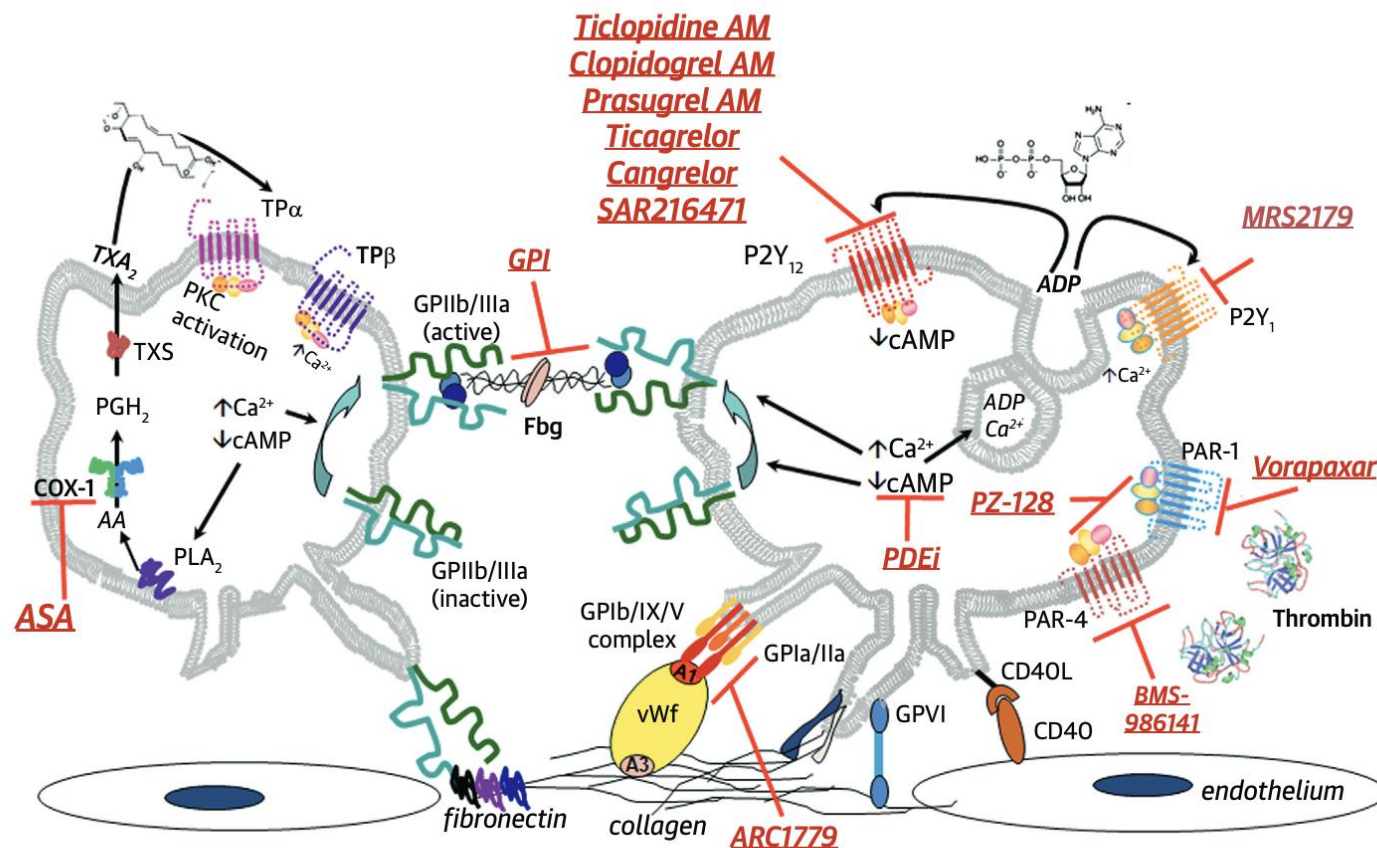
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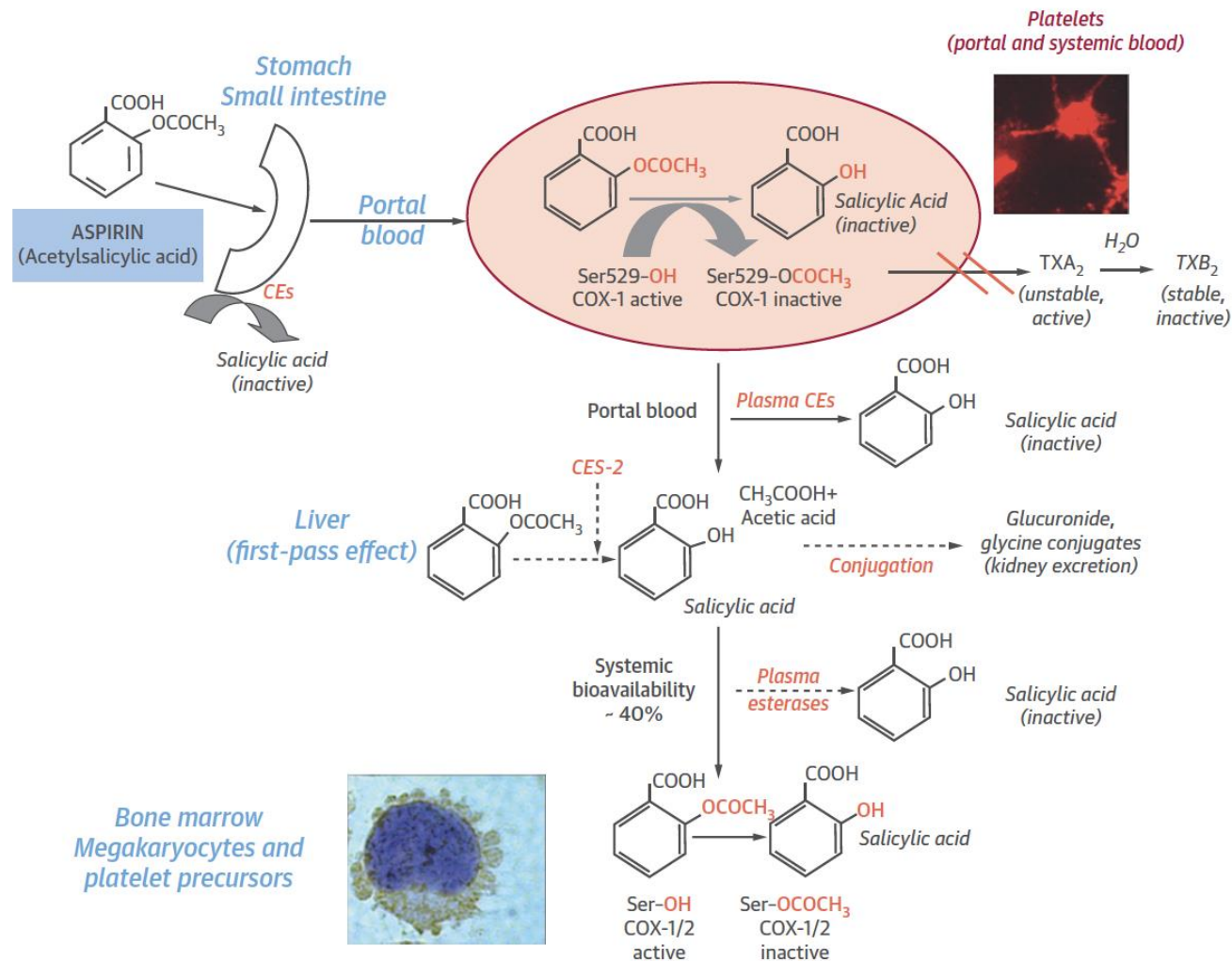


FIGURE 1 Platelet Activation Pathways and Targets of Current and Novel Antiplatelet Drugs



Platelet activation via multiple pathways leads to numerous responses including: secretion of ADP and its binding to P2Y₁ and P2Y₁₂ receptors; changes in the surface membrane supporting thrombin generation and activation of its PAR1 and PAR4 receptors; increased intraplatelet calcium; reduced cAMP concentrations; activation of phospholipases and release of arachidonic acid as a substrate for COX-1 and TXA₂ formation; and final activation of $\alpha 2\beta 3$ (GP IIb/IIIa) leading to fibrinogen binding and platelet aggregation, as well as to amplification signals. Antiplatelet drugs already marketed or under different stages of development (ARC1779, MRS2179, PZ-128, BMS-986141, SAR216471) are shown, together with the targets they interfere with. Adapted from Patrono C, Rocca B. The future of antiplatelet therapy in cardiovascular disease. *Ann Rev Med* 2010;61:49–61. AA = arachidonic acid; ADP = adenosine diphosphate; AM = active metabolite; ASA = aspirin; Ca²⁺ = calcium; cAMP = cyclic adenosine monophosphate; COX = cyclooxygenase; Fbg = fibrinogen; GP = glycoprotein; GPI = GPIIb/IIIa inhibitors; PAR = protease-activated receptor; PDEi = phosphodiesterase inhibitors; PGH₂ = prostaglandin H₂; PKC = protein kinase C; PL = phospholipase; PLA₂ = phospholipase A₂; TP = thromboxane receptor; TX = thromboxane; TXS = thromboxane synthase; vWf = von Willebrand factor.

FIGURE 2 Pharmacokinetics and Pharmacodynamics of Aspirin



Aspirin is absorbed in the stomach and small intestine, exerts its pharmacodynamic effect, that is, the permanent acetylation of a Ser-529 residue of COX-1, already in the portal blood, and is biotransformed to inactive salicylic acid by intestine, plasma, and liver carboxylesterases, mainly the isoenzyme 2. On average, its systemic bioavailability is approximately 50% of the administered dose. Once in the systemic circulation, aspirin reaches bone marrow megakaryocytes and platelet precursors, inhibiting COX-1 and -2. The COX-1-dependent arachidonic acid pathway in platelets generates mainly TXA₂, which amplifies platelet activation by binding to its platelet receptors. TXA₂ is nonenzymatically hydrolyzed to TXB₂, which is biologically inactive but stable, and can be measured in ex vivo assays, or undergoes further hepatic enzymatic biotransformation in vivo. The potential pharmacodynamic drug-drug interaction site is indicated. Adapted from Rocca B, Dragani A, Pagliaccia F. Identifying determinants of variability to tailor aspirin therapy. *Exp Rev Cardiovasc Ther* 2013;11:365-79. CE = carboxylesterase; MK = megakaryocytes; other abbreviations as in [Figure 1](#).

Aspirin (acetylsalicylic acid)

- In the 18th century, the Reverend Edmund Stone wrote about the value of an extract of bark from the willow tree (of the family *Salix*) for alleviating pain and fever.
- The active ingredient was salicin, which is metabolised to salicylic acid in vivo. Sodium salicylate manufactured from salicin proved highly successful in the treatment of rheumatic fever and gout, but it was a gastric irritant.
- In 1897, Felix Hoffman, a chemist at the Bayer Company, whose father developed abdominal pain with sodium salicylate, succeeded in producing acetylsalicylic acid in a form that was chemically stable. The new preparation proved acceptable to his father and paved the way for the production of aspirin.

Morris J. Brown
Clinical Pharmacology

Mechanism of Aspirin

- Aspirin irreversibly inactivates platelet cyclooxygenase-1 and suppresses thromboxane A₂ generation by selectively acetylating a serine residue (Ser-529) close to the catalytic pocket of the enzyme.
- Similarly to other traditional NSAID, aspirin first reversibly binds to a common docking site (Arg-120) within the cyclooxygenase channel.
- This provides the molecular basis of a pharmacodynamic drug-drug interaction with some NSAIDs, notably ibuprofen and naproxen, the co-administration of which may impair the antiplatelet effect of low-dose aspirin.
- Platelet thromboxane A₂ inhibition by low-dose aspirin is cumulative upon repeated daily dosing, and virtually complete (97-100%)
- Upon aspirin withdrawal, re-appearance of thromboxane A₂ biosynthesis displays a 24- to 48-hour delay with full recovery after 7-10 days.
- These unique characteristics reflect the covalent interaction between a short-lived drug (20-min half-life) with its molecular target expressed in two key compartments: peripheral platelets which survive 7-10 days and bone marrow megakaryocytes which daily release ~10% of new platelets.
- Relationship between platelet COX-1 suppression and the inhibition of thromboxane A₂-dependent platelet activation *in vivo* is strikingly non-linear.

Aspirin (acetylsalicylic acid)

Mode of action.

- The anti-inflammatory, analgesic and antipyretic actions of aspirin are those of NSAIDs in general.

The following additional actions are relevant:

- An antiplatelet effect due to permanent inactivation, by acetylation, of COX-1 in platelets, preventing synthesis of thromboxane A_2 .
- Platelets cannot regenerate the enzyme, and the resumption of thromboxane A_2 production is dependent on the entry of new platelets into the circulation (platelet lifespan is 7 days).
- Thus a continuous antiplatelet effect is achieved with low doses.
- Respiratory stimulation is a characteristic of aspirin intoxication and occurs both directly by stimulation of the respiratory centre and indirectly through increased carbon dioxide production.
- Although aspirin in high dose reduces renal tubular reabsorption of uric acid so increasing its elimination, other treatments for hyperuricaemia are preferred.
- Indeed aspirin should be avoided in gout, as low doses inhibit uric acid secretion and on balance its effects on uric acid elimination are adverse.

Dose Adjustments

- According to body weight
- Hepatic impairment
- Renal impairment
- Elderly

Clinical Uses

- The main use of aspirin is as an antiplatelet agent to prevent arterial thrombotic events due to atherosclerosis. It has largely been overtaken by NSAIDs and paracetamol as an analgesic. In high doses, it is used to treat inflammation in Kawasaki disease, in combination with intravenous immunoglobulin, but in other inflammatory illnesses it has now been superseded by other, more effective agents with fewer side-effects.

Pharmacokinetics.

- Aspirin ($t_{1/2}$ 15 min) is well absorbed from the stomach and upper GI tract. Hydrolysis removes the acetyl group, and the resulting salicylate ion is inactivated largely by conjugation with glycine. At low doses, this reaction proceeds by first-order kinetics with a $t_{1/2}$ of about 4 h, but at high doses and in overdose the process becomes saturated, i.e. kinetics become zero order, and most of the drug in the body is present as the salicylate. The challenge in overdose is to remove salicylate.
- Doses of 75–150 mg/day are used to prevent thrombotic vascular occlusion; 300 mg as immediate treatment for myocardial infarction; 300–900 mg every 4–6 h for analgesia.

Side Effects

- Gastrointestinal effects are those of NSAIDs in general. Effects particularly associated with aspirin are:
- *Salicylism* (the symptoms of an excessive dose): tinnitus and hearing difficulty, dizziness, headache and confusion.
- *Allergy* . Aspirin is a common cause of allergic or pseudoallergic symptoms and signs. Patients exhibit severe rhinitis, urticaria, angioedema, asthma or shock. Those who already suffer from recurrent urticaria, nasal polyps or asthma are more susceptible.
- *Reye's syndrome* . Epidemiological evidence relates aspirin use to the development of the rare Reye's syndrome (encephalopathy, liver injury) in children recovering from febrile viral infections. Consequently, aspirin should not be given to children younger than 15 years of age.

Acute overdose.

- A moderate overdose (plasma salicylate 500–750 mg/L) will cause nausea, vomiting, epigastric discomfort, tinnitus, deafness, sweating, pyrexia, restlessness, tachypnoea and hypokalaemia. A large overdose (plasma salicylate concentration above 750 mg/L) may result in pulmonary oedema, convulsions and coma, with severe dehydration and ketosis. Bleeding is unusual, despite the antiplatelet effect of aspirin.
- Metabolic acidosis suggests severe poisoning

Heparin

- a parenteral anticoagulant widely used in clinical medicine.
- discovered in 1916 and was approved by the FDA in 1939.
- It is a glycosaminoglycan composed of chains of alternating residues of D-glucosamine and an uronic acid; its molecular weight ranges from 5,000—30,000 daltons, with a mean of 15,000 daltons (about 50 monosaccharide chains).
- Commercially derived from porcine or bovine tissue, and is available as the calcium or sodium salt.
- Full therapeutic doses of heparin prolong clotting time; however, bleeding time is usually unaffected by heparin. The response to heparin is variable since only about one-third of the molecules have anticoagulant activity, and its activity and clearance is dependent upon the length and size of the heparin molecules.
- Compared with low molecular weight heparins, unfractionated heparin produces a less predictable anticoagulant response due primarily to its reduced bioavailability after subcutaneous administration of low doses, its dose-dependent clearance, and differences among patients in the nonspecific binding of heparin to proteins and cells.

Heparin- Indications

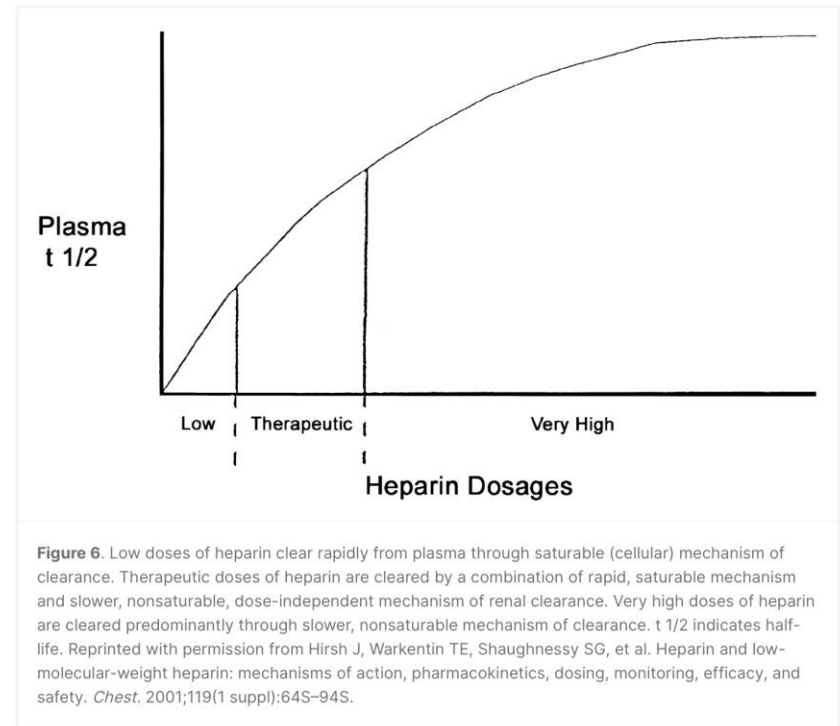
- acute myocardial infarction
- NSTEMI/ STEMI
- Antiphospholipid antibody syndrome/ arterial thromboembolism
- deep venous thrombosis (DVT) or prophylaxis
- thromboembolic complications of AF
- pulmonary embolism
- percutaneous coronary intervention (PCI)
- thrombosis prophylaxis

Mechanism of Action- Heparin

- Exerts its anticoagulant action by accelerating the activity of antithrombin III (ATIII) to inactivate thrombin
- however, heparin does not lyse existing clots.
- Approximately one-third of heparin molecules contain a unique pentasaccharide sequence with high-affinity binding to ATIII. The interaction of heparin with ATIII produces a conformational change in ATIII, which accelerates the ability of ATIII to inactivate thrombin (factor IIa), factor Xa, and factor IXa.
- Of these enzymes, thrombin is the most sensitive to inhibition by heparin/ATIII.
- The inability of heparin to inactivate surface-bound thrombin and factor Xa may explain its limited efficacy in patients with unstable angina, high-risk coronary angioplasty, and coronary thrombolysis.
- The variability of anticoagulant response in individuals given fixed doses of heparin is thought to be due to differences between patients in their plasma concentrations of neutralizing plasma proteins and/or heparin-binding proteins.
- Patients with acquired antithrombin deficiency (< 25% normal concentration) may not respond to heparin.
- Protamine as the antidote

Pharmacokinetics- Heparin

- Given parenterally, either intravenously or subcutaneously.
- Heparin is highly bound to antithrombin, fibrinogens, globulins, serum proteases, and lipoproteins.
- does not undergo enzymatic degradation.
- primarily cleared from the circulation by liver and reticuloendothelial cells mediated uptake into extravascular space.
- Heparin undergoes biphasic clearance, first by a rapid, saturable, zero-order process, then via slower first order elimination. In the zero-order phase, heparin is bound to the surface of cells (e.g., proteins, endothelial cells, macrophages) where it is internalized and depolymerized.
- Low doses of heparin are cleared mostly by a saturable, rapid, zero-order process. Slower first order elimination usually occurs with very high doses of heparin and is dependent on renal function.
- Plasma half-life is dose-dependent and ranges from 0.5 to 2 hours.



Heparin Administration

Intravenous Route (unfractionated heparin)

- Peak plasma concentration and onset of action are achieved immediately after intravenous administration.

Subcutaneous Route (LMWH, e.g. enoxaparin, tinzaparin)

- Administration of low or moderate doses of subcutaneous heparin reduces the plasma recovery. However, plasma recovery is almost complete with use of high therapeutic doses of subcutaneous heparin (more than 35,000 units every 24 hours).
- After subcutaneous administration, approximately 22% to 40% of the administered dose is absorbed systemically as determined by anti-factor Xa assay. Subcutaneous heparin concentrations reach steady-state between 4 and 10 hours.

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HEPARIN PROTOCOL
(FOR REFERENCE ONLY)

For Pressure Lines: e.g. swan ganz catheter, IABP pressure port
Heparin Concentration: 500 units in 250 ml Normal Saline (Soft bag)
i.e. 2 units heparin / ml NS

For Therapeutic Treatment

Heparin Concentration: 5000 units in 100 ml Normal Saline
i.e. 50 units heparin / ml NS

* Check PT/APTT Q4H to Q8H

To maintain APTT between 50-75 seconds:-

APTT <50	heparin 1500u IV bolus, then increase drip rate by 100u / hr.
APTT 50-75	no change.
APTT 76-90	decrease drip rate by 100u / hr.
APTT 91-110	stop heparin for 1 hour, then decrease drip rate by 100u / hr.
APTT >110	stop heparin for 1.5 hour, then decrease drip rate by 200u / hr.

Important notes

- Heparin concentration for pressure line and therapeutic treatment are standard practice in Coronary Care Unit (CCU) QMH.
- The maintenance of APTT level may be varied, please confirm with the responsible doctor for heparin adjustment.
- The frequency of PT/APTT checking should be adjusted according to clinical condition of patient.

Antiplatelet Therapies

- Multiple pathways contribute to platelet activation and aggregation, and although pharmacological interference with these pathways reduces the risk of atherothrombotic complications, it is also associated with an increased risk of bleeding.
- It is important to emphasize that the thromboxane (TX) A₂, adenosine diphosphate (ADP), and thrombin-activated pathways transduce independent signals of platelet activation and represent nonredundant targets for its pharmacological modulation.
- This is reflected by the additive nature of the effects of combined antiplatelet therapy

P2Y12 Inhibitors

- Oral inhibitors of the platelet ADP receptor, P2Y12, include the thienopyridines (ticlopidine, clopidogrel, and prasugrel) and ticagrelor
- Thienopyridines are prodrugs, generating short-lived active metabolites that irreversibly inactivate the receptor and consequently inhibit ADP-induced platelet activation.
- When added to COX-1 suppression by low-dose aspirin, P2Y12 blockade by clopidogrel produces an additional 10% to 20% relative risk reduction of major vascular events in high-risk patients.
- This relatively modest benefit may reflect the low degree of P2Y12 inactivation achieved by clopidogrel in most patients.

Ticagrelor

- Ticagrelor is an adenosine triphosphate analogue.
- It directly and reversibly binds the P2Y₁₂ receptor, acting as an allosteric antagonist that noncompetitively prevents ADP-induced P2Y₁₂ activation.
- Ticagrelor inhibits adenosine diphosphate-dependent platelet aggregation more rapidly and profoundly than clopidogrel (~80% inhibition within 2 hours), and recovery occurs within 3-4 days after withdrawal.
- A faster and more complete P2Y₁₂ blockade by prasugrel or ticagrelor produced additional benefit versus clopidogrel in ACS, supporting the clinical relevance of effectively targeting 2 nonredundant platelet signaling pathways.

Ticagrelor

- Ticagrelor has a half-life of 7-12 hours and ~36% bioavailability.
- Ticagrelor is biotransformed by CYP3A4/A5 to a main active and an inactive metabolite. AR-C124910XX is equipotent to ticagrelor, has similar pharmacokinetics, and contributes approximately 30% of the antiplatelet effect.
- P-glycoprotein or CYP2C19 polymorphisms do not affect ticagrelor pharmacokinetics.
- Clinically relevant drug-drug interactions may occur with strong CYP3A4 inhibitors or inducers, which are contraindicated.
- A clinically relevant P-glycoprotein-related interaction might occur with digoxin, due to its narrow therapeutic window

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Ticagrelor versus Clopidogrel in Patients with Acute Coronary Syndromes

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ABSTRACT

BACKGROUND

Ticagrelor is an oral, reversible, direct-acting inhibitor of the adenosine diphosphate receptor P2Y₁₂ that has a more rapid onset and more pronounced platelet inhibition than clopidogrel.

METHODS

In this multicenter, double-blind, randomized trial, we compared ticagrelor (180-mg loading dose, 90 mg twice daily thereafter) and clopidogrel (300-to-600-mg loading dose, 75 mg daily thereafter) for the prevention of cardiovascular events in 18,624 patients admitted to the hospital with an acute coronary syndrome, with or without ST-segment elevation.

RESULTS

At 12 months, the primary end point — a composite of death from vascular causes, myocardial infarction, or stroke — had occurred in 9.8% of patients receiving ticagrelor as compared with 11.7% of those receiving clopidogrel (hazard ratio, 0.84; 95% confidence interval [CI], 0.77 to 0.92; $P<0.001$). Predefined hierarchical testing of secondary end points showed significant differences in the rates of other composite end points, as well as myocardial infarction alone (5.8% in the ticagrelor group vs. 6.9% in the clopidogrel group, $P=0.005$) and death from vascular causes (4.0% vs. 5.1%, $P=0.001$) but not stroke alone (1.5% vs. 1.3%, $P=0.22$). The rate of death from any cause was also reduced with ticagrelor (4.5%, vs. 5.9% with clopidogrel; $P<0.001$). No significant difference in the rates of major bleeding was found between the ticagrelor and clopidogrel groups (11.6% and 11.2%, respectively; $P=0.43$), but ticagrelor was associated with a higher rate of major bleeding not related to coronary-artery bypass grafting (4.5% vs. 3.8%, $P=0.03$), including more instances of fatal intracranial bleeding and fewer of fatal bleeding of other types.

CONCLUSIONS

In patients who have an acute coronary syndrome with or without ST-segment elevation, treatment with ticagrelor as compared with clopidogrel significantly reduced the rate of death from vascular causes, myocardial infarction, or stroke without an increase in the rate of overall major bleeding but with an increase in the rate of non-procedure-related bleeding. (ClinicalTrials.gov number, NCT00391872.)

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*The Study of Platelet Inhibition and Patient Outcomes (PLATO) investigators are listed in the Appendix and the Supplementary Appendix, available with the full text of this article at NEJM.org.

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TABLE 1 Main Properties of P2Y₁₂ Inhibitors				
	Clopidogrel	Prasugrel	Ticagrelor	Cangrelor
Bioavailability	50%	80%	36%	100%
Half-life (active metabolite)	30-60 min	Distribution half-life 30-60 min; elimination half-life 2-15 h	7-9 h	3-6 min
Binding reversibility	Irreversible	Irreversible	Reversible	Reversible
Onset of action	2-6 h	30 min	30 min	2 min
Frequency of administration	Once daily	Once daily	Twice daily	Intravenous infusion
Duration of effect	3-10 days	7-10 days	3-5 days	1-2 h
Antidote	No	No	No	No
Clinical indication	PCI, ACS, ACS-PCI, and SCAD	ACS-PCI	ACS, ACS-PCI	PCI, bridging in patient at high ischemic risk who is undergoing an invasive procedure
Discontinuation before nonacute surgery	At least 5 days	At least 7 days	At least 3 days	1 h
ACS = acute coronary syndromes; PCI = percutaneous coronary intervention; SCAD = stable coronary artery disease.				

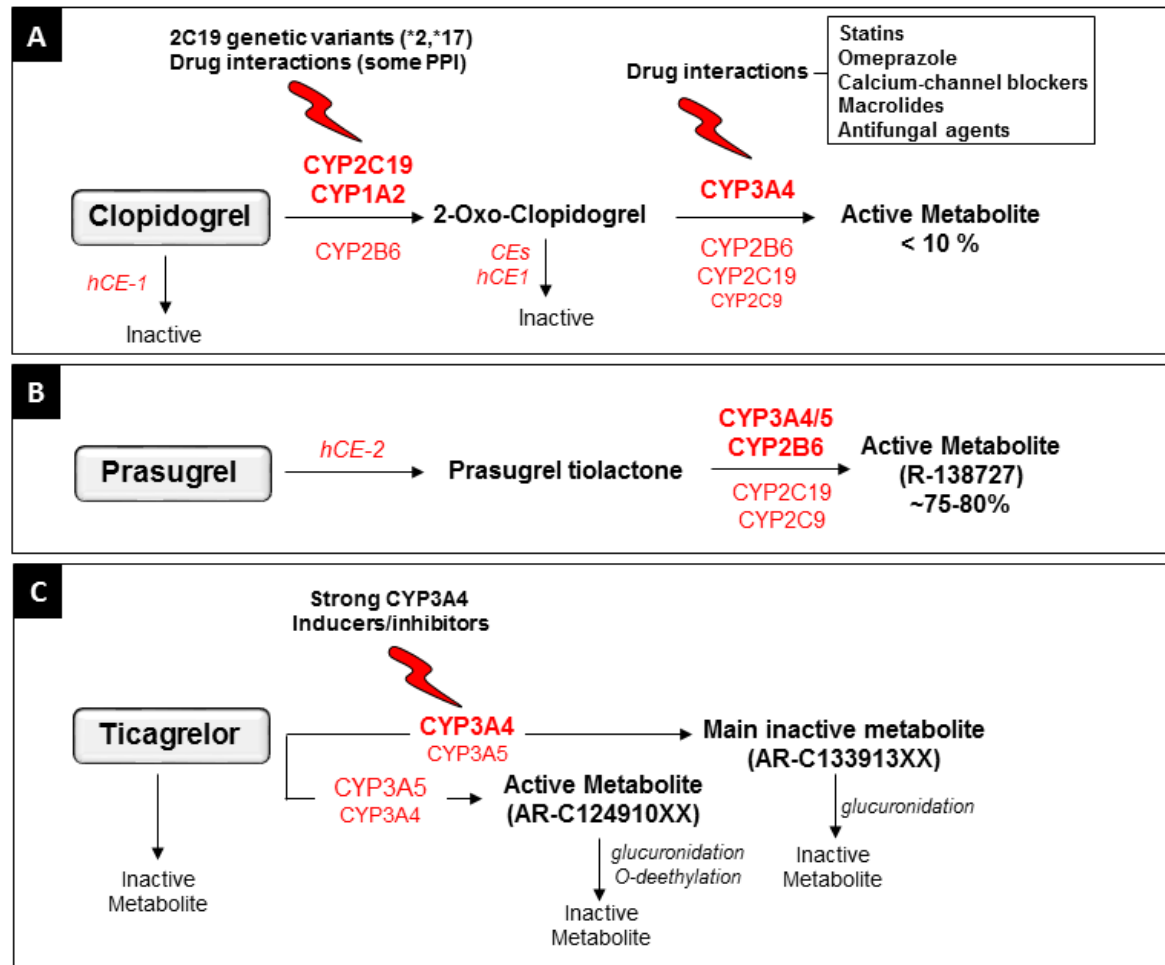
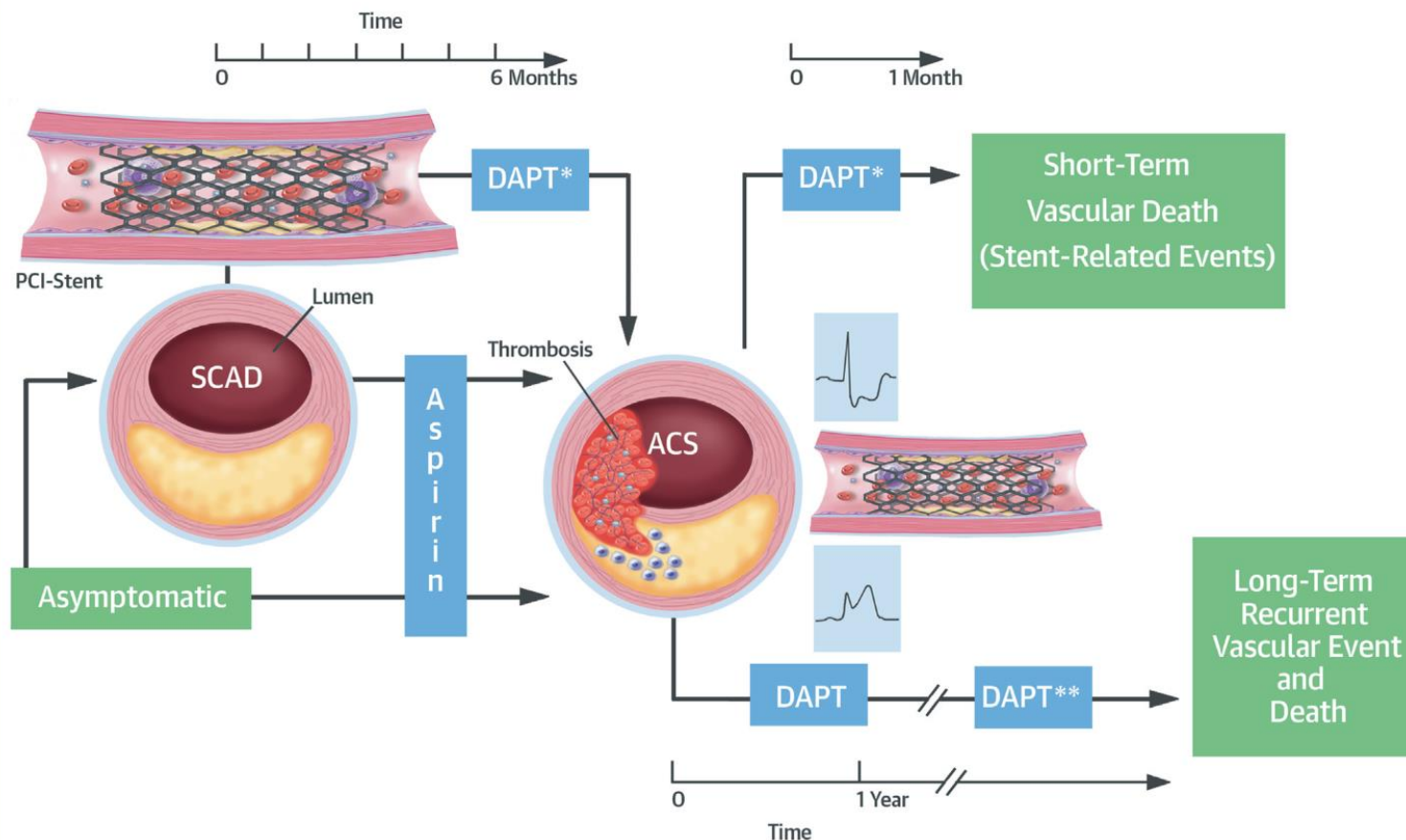


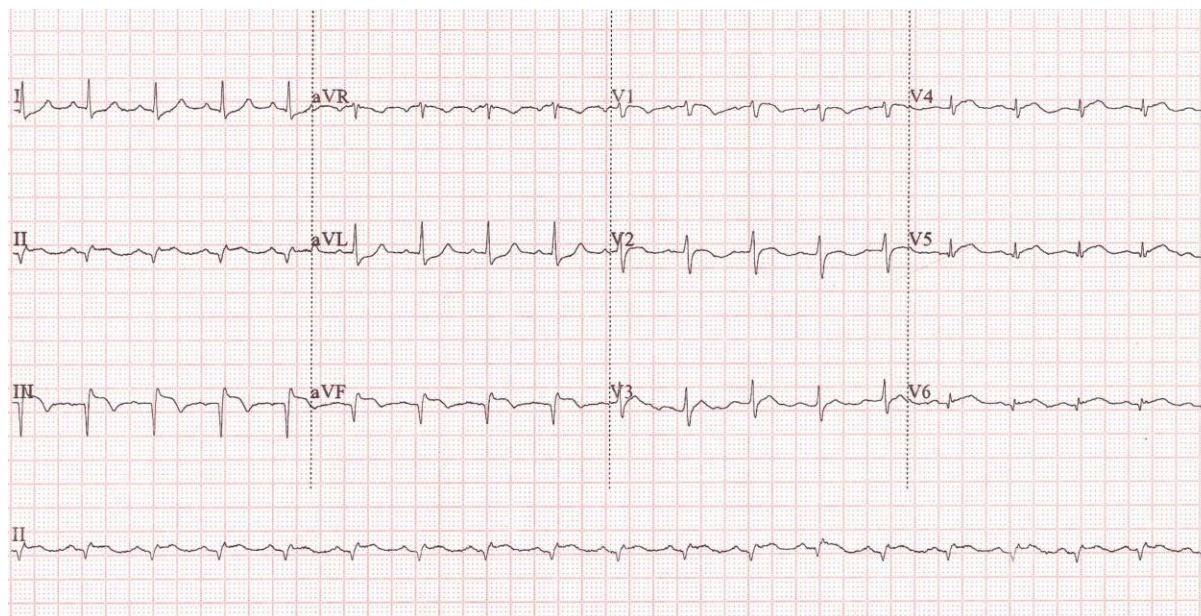
Figure 1

CENTRAL ILLUSTRATION Coronary Atherothrombosis and its Prevention

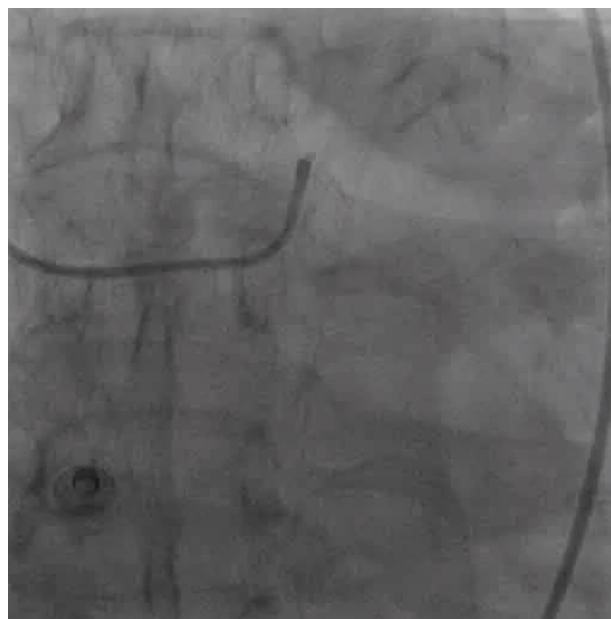
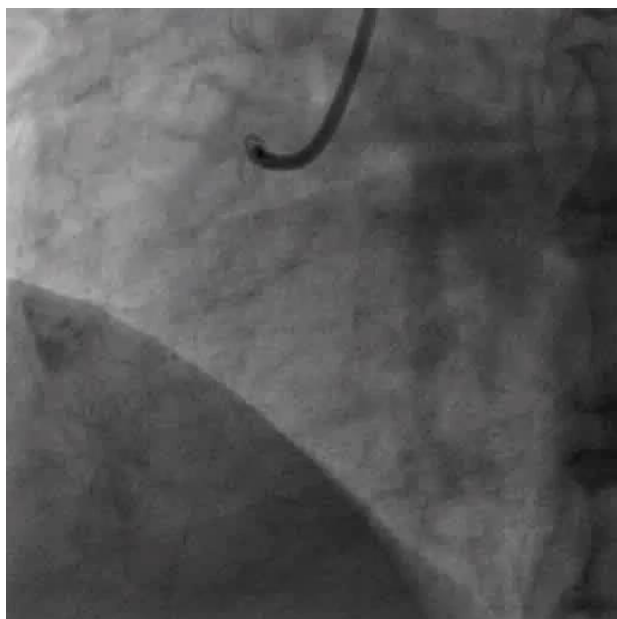


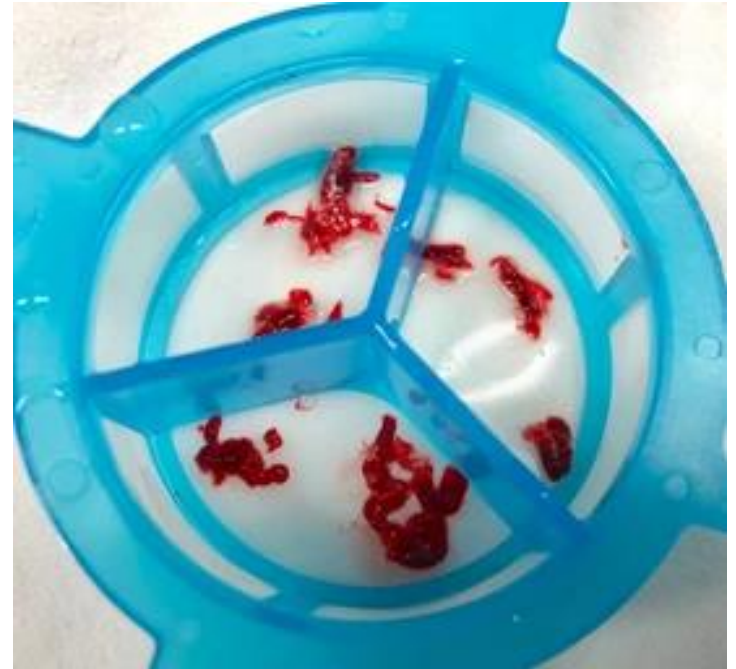
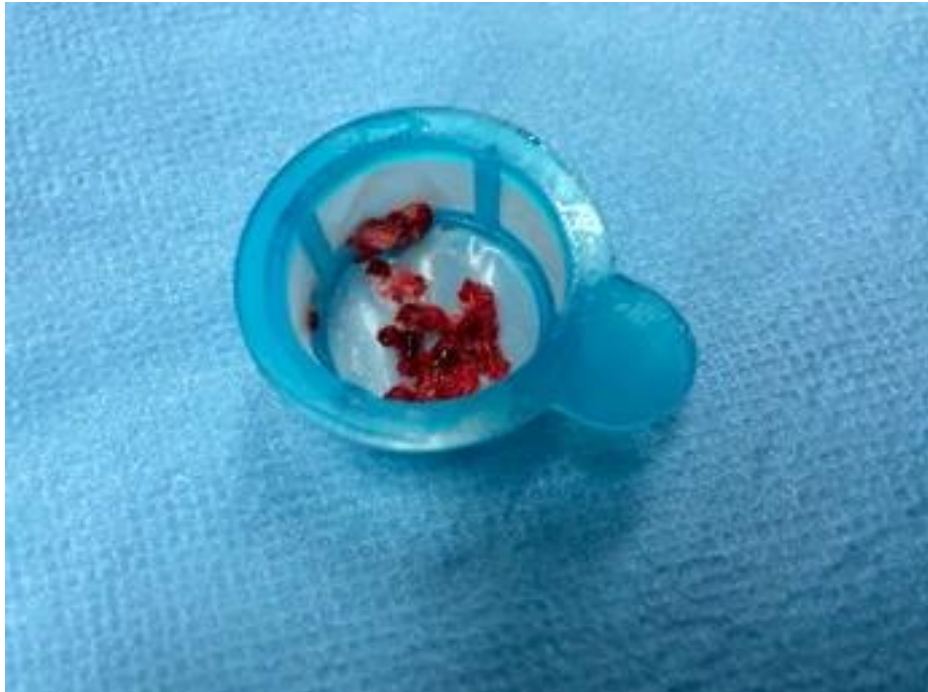
Patrono, C. et al. *J Am Coll Cardiol.* 2017;70(14):1760-76.

Different pathways in the natural history of coronary atherothrombosis and evidence-based interventions with antiplatelet agents for its treatment and prevention (both primary and secondary) that are reviewed throughout the paper. The time scales are intended to represent the timeframe within which particular combinations of antiplatelet agents were shown to be effective in reducing any given outcome illustrated in the figure. ACS = acute coronary syndrome; DAPT = dual antiplatelet therapy with aspirin + clopidogrel/ticagrelor/prasugrel; DAPT* = dual antiplatelet therapy with aspirin + clopidogrel; DAPT** = dual antiplatelet therapy with aspirin + ticagrelor 60 mg or aspirin + vorapaxar; PCI = percutaneous coronary intervention; SCAD = stable coronary artery disease.



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Photos from worldwide web

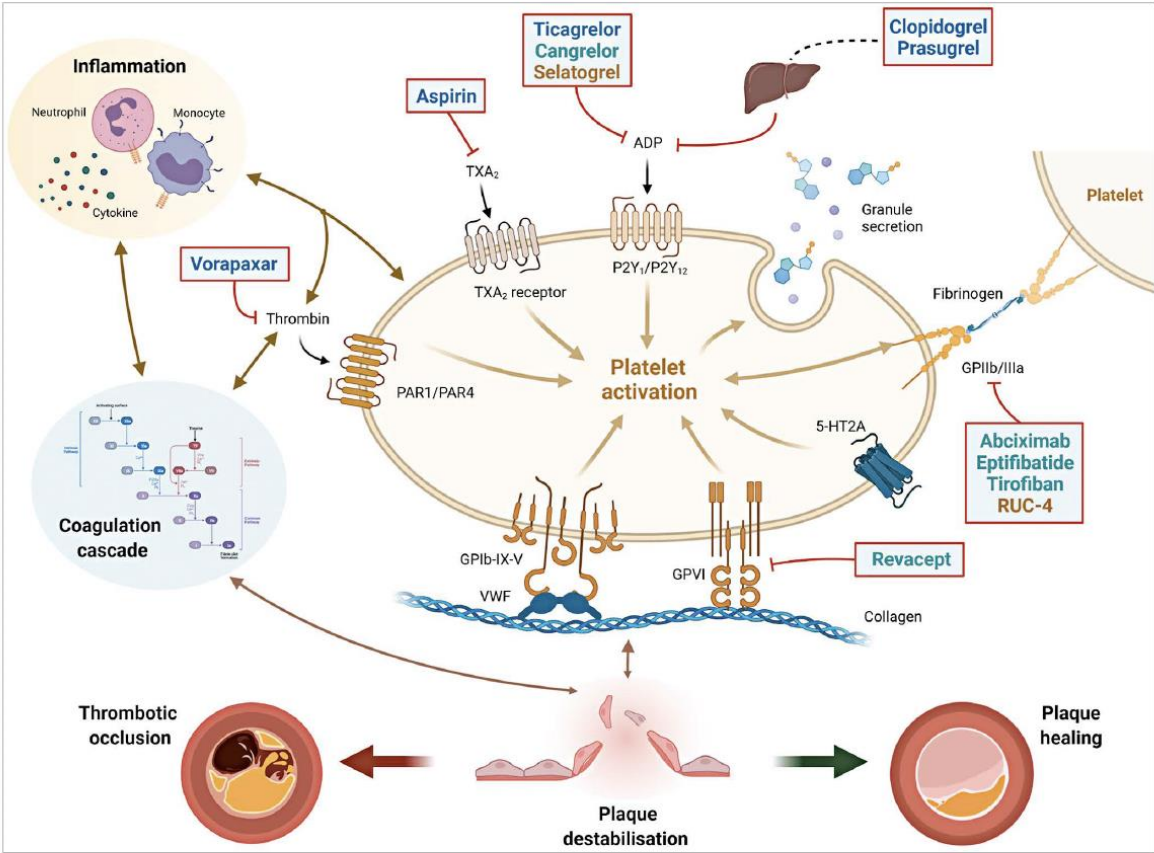


Figure 1. Interplay between platelets, coagulation and inflammation in atherothrombosis and sites of action of antiplatelet agents. The interplay between platelets, coagulation and inflammation is a key modulator of atherosclerosis and its thrombotic complications. When plaque destabilisation occurs, due to plaque rupture or erosion, platelets and coagulation factors, as well as subsequent inflammatory processes, it may lead to either a thrombotic occlusion of the coronary artery (leading to acute coronary syndromes) or plaque healing (favouring plaque progression with stable coronary artery disease). Initial platelet tethering is mediated by the interaction between the complex glycoprotein (GP) Ib-LX-V and Von Willebrand factor (vWF) and by other collagen receptors present on the platelet surface such as GPVI. Thrombin is a key linking factor between platelet and coagulation cascade. Sites of action of common antiplatelet agents are reported: aspirin inhibits thromboxane A₂ (TXA₂); clopidogrel, prasugrel, ticagrelor and cangrelor are inhibitors of the ADP P₂Y₁₂ receptor. Clopidogrel and prasugrel require hepatic activation. Vorapaxar is a thrombin receptor inhibitor (protease-activated receptor; PAR-1); abciximab, eptifibatide, tirofiban and RUC-4 are GPIIb/IIIa receptor inhibitors. Revacept is a competitive antagonist of the collagen-GPVI signalling. Routes of administration: Blue=oral; Green=intravenous; Yellow=subcutaneous/intramuscular. ADP: adenosine diphosphate; GP: glycoprotein; PAR-1: platelet protease-activated receptor-1; TXA₂: thromboxane A₂; vWF: von Willebrand factor; 5HT_{2A}: serotonin receptor.



Glycoprotein (GP)IIb/IIIa inhibitors.

- Intravenous GPIIb/IIIa inhibitors (GPI) prevent fibrinogen binding to activated GPIIb/IIIa receptor and the consequent formation of fibrinogen bridges between platelets, which is the final common pathway of platelet aggregation.
- abciximab, eptifibatide, and tirofiban.
- Abciximab, is the chimeric Fab-fragment of the murine monoclonal antibody, 7E3, and a non-competitive receptor blocker.
- Eptifibatide, a cyclic heptapeptide, and tirofiban, a non-peptide mimetic molecule, are modeled on KDG (Lys-Gly-Asp) and RGD (Arg-Gly-Asp) sequences, respectively, and competitively inhibit fibrinogen binding.
- Their effect on platelet aggregation is closely related to plasma concentrations. Owing to their short plasma half-lives, bolus and then continuous infusion are needed.
- Clinically-relevant PD interactions between GPIs, P2Y₁₂ blockers or fibrinolytic agents have not been reported.



Dipyridamole / cilostazol

- Dipyridamole and cilostazol are (PDE) inhibitors, with vasodilator and antiplatelet effects.
- They increase intraplatelet cAMP and cGMP levels and reduce cellular adenosine uptake into red blood cells, platelets and endothelial cells.
- The absorption of dipyridamole from conventional formulations is quite variable with low systemic bioavailability. A modified, extended-release formulation with improved bioavailability has been developed with a terminal half-life of 10 h and twice-daily dosing.
- The fixed combination of low-dose aspirin and extended-release dipyridamole is used in the secondary prevention of stroke. Headache is a common adverse event limiting compliance.

Direct Oral Anticoagulants



Table 1. Indications for DOAC Prescribing ([Table view](#))

DOACs	Indications
Dabigatran	FDA-approved indications
	Stroke prevention in NVAF
	Treatment of deep vein thrombosis and pulmonary embolism
	Prevention of recurrent deep vein thrombosis and pulmonary embolism
	Prevention of thromboembolism after total hip replacement
	Off-label indications
	Prevention of thromboembolism after total knee replacement
	Prevention of thromboembolism after PCI with NVAF
Rivaroxaban	FDA-approved indications
	Stroke prevention in NVAF
	Treatment of deep vein thrombosis and pulmonary embolism
	Prevention of recurrent deep vein thrombosis and pulmonary embolism
	Prevention of thromboembolism after total knee replacement and after total hip replacement
	Prevention of thromboembolism in hospitalized acutely ill medical patients
	Prevention of major cardiovascular events in patients with chronic CAD/peripheral artery disease
	Off-label indications
Apixaban	Prevention of thromboembolism after PCI with NVAF
	FDA-approved indications
	Stroke prevention in NVAF
	Prevention of thromboembolism after total knee replacement and after total hip replacement
	Treatment of deep vein thrombosis and pulmonary embolism
	Prevention of recurrent deep vein thrombosis and pulmonary embolism
	Off-label indications
	Treatment of heparin-induced thrombocytopenia;
Edoxaban	Prevention and treatment of cancer-associated deep vein thrombosis
	Prevention of thromboembolism in hospitalized acutely ill medical patients
	Prevention of thromboembolism after PCI with NVAF
	FDA-approved indications
	Stroke prevention in NVAF
	Treatment of deep vein thrombosis and pulmonary embolism
	Off-label indications
	Prevention and treatment of cancer associated deep vein thrombosis
Betrixaban	Prevention of thromboembolism after total knee replacement and after total hip replacement
	Prevention of thromboembolism after PCI with peripheral artery disease
	FDA-approved indications
Betrixaban	Prevention of deep vein thrombosis and pulmonary embolism in adults hospitalized for an acute medical illness

Direct Oral Anticoagulants

Table I New anticoagulant drugs, approved or under evaluation for prevention of systemic embolism or stroke in patients with non-valvular atrial fibrillation

	Dabigatran	Apixaban	Edoxaban ^a	Rivaroxaban
Action	Direct thrombin inhibitor	Activated factor Xa inhibitor	Activated factor Xa inhibitor	Activated factor Xa inhibitor
Dose	150 mg bid 110 mg bid	5 mg bid 2.5 mg bid	60 mg qd 30 mg qd 15 mg qd	20 mg qd 15 mg qd
Phase 3 clinical trial	RE-LY ³	ARISTOTLE ⁴ AVERROES ⁴	ENGAGE-AF ⁵	ROCKET-AF ⁶

^aNo EMA approval yet. Needs update after finalization of SmPC.

bid, twice daily; qd, once daily.

See further Tables and text for discussion on dose considerations.

Hatching, as (being) studied in Phase 3 clinical trial; not yet approved by EMA.

Agent	Stroke Prevention Atrial Fibrillation	Dosing Adjustments*
Apixaban	5 mg orally twice daily	Dose reduction to 2.5 mg orally twice daily if the patient meets at least 2 of the following 3 characteristics: ■ Age $\geq$80 years ■ Actual body weight $\leq$60 kg and/or ■ Serum creatinine $\geq$1.5 mg/dL Patients with ESKD receiving hemodialysis were not enrolled in clinical trials. However, the prescribing information suggests no dosing adjustment for patients with ESKD, unless they have additional dose reduction characteristics.
Dabigatran	150 mg orally twice daily 110 mg orally twice daily	Dose reduction to 75 mg orally twice daily if the CrCl (estimated using actual body weight) is 15–30 mL/min. Dose reduction is not recommended for patients with ESKD in the 2019 ACC/AHA/HRS AF guideline focused update. [†] The FDA has not approved this dose for use in AF in the United States. In the European label, a twice-daily dose of 110 mg is recommended for patients age $\geq$ 80 years and “for consideration” for those age 75–80 years.
Edoxaban	60 mg orally once daily	The 60-mg once-daily dose is for patients with a CrCl (estimated using actual body weight) of 51–95 mL/min (not recommended for patients with a CrCl $>$ 95 mL/min). Dose reduction to 30 mg orally once daily if the CrCl is 15–50 mL/min. Patients with ESKD were not enrolled in clinical trials, and the prescribing information provides no dosing recommendations for patients with ESKD.
Rivaroxaban	20 mg orally once daily 15 mg orally once daily	Dose reduction to 15 mg orally once daily with the evening meal for patients with CrCl (estimated using actual body weight) $\leq$ 50 mL/min. Patients with ESKD were not enrolled in clinical trials. Dose reduction not recommended for patients with ESKD in the 2019 ACC/AHA/HRS AF guideline focused update. [†] This is the dose that was studied in PIONEER AF-PCI, with adjunctive use of a P2Y ₁₂ i. This dose has not been approved for stroke prevention in AF for those with a CrCl $>$ 50 mL/min without concomitant P2Y ₁₂ i use. When this lower, off-label dosing strategy is used in those with CrCl $<$ 50, the dose should be adjusted to 10 mg.
VKA	When used with APT INR 2.0–3.0 [‡]	NA

*Dosing information in this table does not take drug–drug interactions into consideration. The reader is encouraged to review the specific drug prescribing information.

[†]Although the prescribing information provides dosing recommendations for patients with ESKD, the 2019 ACC/AHA/HRS AF guideline focused update recommends apixaban or warfarin for patients with ESKD. Dabigatran, rivaroxaban, and edoxaban are considered Class III (no benefit) in this population.

[‡]Reasonable to aim for the lower end of the target INR range (i.e., 2.0–2.5). Monitor INR more frequently.

ACC = American College of Cardiology; AF = atrial fibrillation; AHA = American Heart Association; APT = antiplatelet therapy; CrCl = creatinine clearance; ESKD = end-stage kidney disease; FDA = U.S. Food and Drug Administration; HRS = Heart Rhythm Society; INR = international normalized ratio; NA = not applicable; P2Y₁₂i = P2Y₁₂ inhibitor; PIONEER AF-PCI = An Open-Label, Randomized, Controlled, Multicenter Study Exploring Two Treatment Strategies of Rivaroxaban and a Dose-Adjusted Oral Vitamin K Antagonist Treatment Strategy in Subjects With Atrial Fibrillation Who Undergo Percutaneous Coronary Intervention; VKA = vitamin K antagonist.

Agent	VTE Initial Treatment	VTE Secondary Prevention after Initial Therapy	Dosing Adjustments*
Apixaban	10 mg orally twice daily for the first 7 days of therapy followed by 5 mg orally twice daily.	After ≥6 months of initial therapy, either 5 mg orally twice daily or 2.5 mg orally twice daily can be considered. [†]	Patients with ESKD receiving hemodialysis were not enrolled in clinical trials. However, the prescribing information states that no dose adjustment is necessary for patients with renal impairment, including those with ESKD.
Dabigatran	150 mg orally twice daily when preceded by 5-10 days of parenteral AC. [‡]	150 mg orally twice daily. [§]	Patients with severe renal impairment (a CrCl of ≤30 mL/min) and with ESKD receiving hemodialysis were not enrolled in clinical trials. The prescribing information makes no recommendations for dosing in this population.
Edoxaban	60 mg orally once daily when preceded by at least 5-10 days of parenteral AC. [‡]	60 mg orally once daily.	Dose reduction to 30 mg once daily for patients with a CrCl (estimated using actual body weight) of 15-50 mL/min or body weight ≤60 kg.
Rivaroxaban	15 mg orally twice daily with food for the first 21 days followed by 20 mg daily with food.	After ≥6 months of initial therapy, either 20 mg orally daily with food or 10 mg orally daily with or without food can be considered. [†]	Patients with a CrCl of <30 mL/min were excluded from clinical trials. Avoid use in patients with a CrCl of <15 mL/min.
VKA	When used with APT: INR 2.0-2.5 ; bridging with parenteral heparin initially.	When used with APT: Consider INR 2.0-2.5.	NA
Dalteparin	In the setting of cancer: 200 units/kg subcutaneously once daily for 1 month, then 150 IU/kg subcutaneously once daily (months 2-6) for extended treatment.	In the setting of cancer: Not FDA-approved for this indication, but use is consistent with NCCN recommendations.	For patients with a CrCl of <30 mL/min, the prescribing information recommends monitoring anti-Factor Xa levels with a target peak level (4-6 hours post-dose) of 0.5-1.5 IU/mL. Patients with ESKD were excluded from clinical trials.
Enoxaparin	In the setting of cancer: 1 mg/kg twice daily or 1.5 mg/kg once daily, subcutaneously. ^{¶**}	In the setting of cancer: Not FDA-approved for this indication, but use is consistent with NCCN recommendations ^{¶**}	Patients with a CrCl of <30 mL/min were excluded from clinical trials. However, the prescribing information recommends a dose reduction to 1 mg/kg subcutaneously once daily for patients with a CrCl (estimated using actual body weight) of <30 mL/min).

*Dosing information in this table does not take drug-drug interactions into consideration. The reader is encouraged to review the specific drug prescribing information.

†Reduced-dose rivaroxaban (10 mg daily) and apixaban (2.5 mg twice daily) can be considered for secondary prevention of VTE after 6 months of initial treatment.

‡Initial treatment with unfractionated heparin, LMWH, or fondaparinux recommended.

§Dabigatran 110 mg twice daily is approved for use in DVT/PE treatment outside of the United States.

||Monitor INR more frequently.

¶Long-term treatment with enoxaparin at this dose has not been tested in cancer patients.

**Agent and dosing supported by the NCCN Clinical Practice Guidelines in Oncology for Cancer-Associated Venous Thromboembolic Disease (Version 1.2019).

***Among patients without cancer, enoxaparin is approved for DVT and is also used extensively off-label for treatment of PE

AC = anticoagulation; APT= antiplatelet therapy; CrCl = creatinine clearance; DVT = deep vein thrombosis; ESKD = end-stage kidney disease; FDA = U.S. Food and Drug Administration; INR = international normalized ratio; LMWH = low-molecular-weight heparin; NA = not applicable; NCCN = National Comprehensive Cancer Network; PE = pulmonary embolism; VKA = vitamin K antagonist.

2020 ACC Expert Consensus Decision Pathway for Anticoagulant and Antiplatelet Therapy in Patients With AF or VTE Undergoing PCI or With Atherosclerotic CV Disease

J Am Coll Cardiol 2021

Direct Oral Anticoagulants

Table 4 Absorption and metabolism of the different NOACs

	Dabigatran	Apixaban	Edoxaban ^a	Rivaroxaban
Bio-availability	3–7%	50%	62% ¹⁷	66% without food Almost 100% with food
Prodrug	Yes	No	No	No
Clearance non-renal/renal of absorbed dose (if normal renal function; see also Section 8)	20%/80%	73%/27% ¹⁸	50%/50% ⁹	65%/35%
Liver metabolism: CYP3A4 involved	No	Yes (elimination; minor CYP3A4 contribution) ¹⁹	Minimal (<4% of elimination)	Yes (elimination)
Absorption with food	No effect	No effect	6–22% more ²⁰	+39% more ²¹
Intake with food recommended?	No	No	No official recommendation yet	Mandatory
Absorption with H2B/PPI	– 12–30% ^{22–24}	No effect	No effect	No effect ^{21,25}
Asian ethnicity	+25% ²⁴	No effect	No effect ²⁰	No effect
GI tolerability	Dyspepsia 5–10%	No problem	No problem	No problem
Elimination half-life	12–17 h ²³	12 h	9–11 h ⁹	5–9 h (young) 11–13 h (elderly)

^aNo EMA approval yet. Needs update after finalization of SmPC.
H2B, H2-blocker; PPI, proton-pump inhibitor; GI, gastro-intestinal.

Direct Oral Anticoagulants

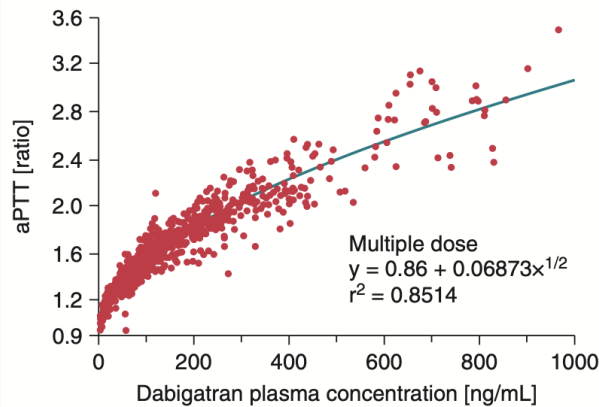


Figure 3 Curvilinear relation between aPTT and dabigatran plasma levels. From van Ryn *et al.*,¹² with permission.

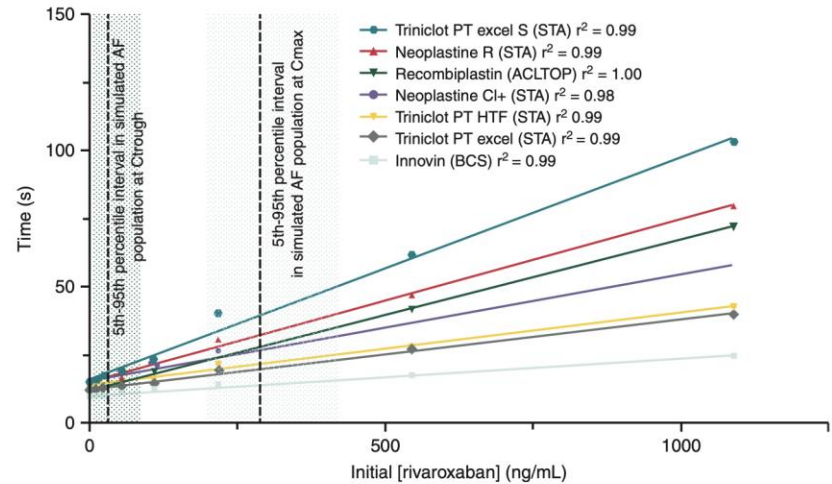


Figure 4 Relation between PT and FXa inhibitor (rivaroxaban) plasma levels. But the sensitivity is dependent on the reagent used. The figure shows data for rivaroxaban. The slopes will vary for other FXa inhibitors and reagents. Moreover, as in Figure 3, there is a high variation of measured values at different concentrations. It is clear that only qualitative information can be gained, without precise measurement of the anticoagulant effect. From Douxils *et al.*¹⁵ with permission.

Direct Oral Anticoagulants

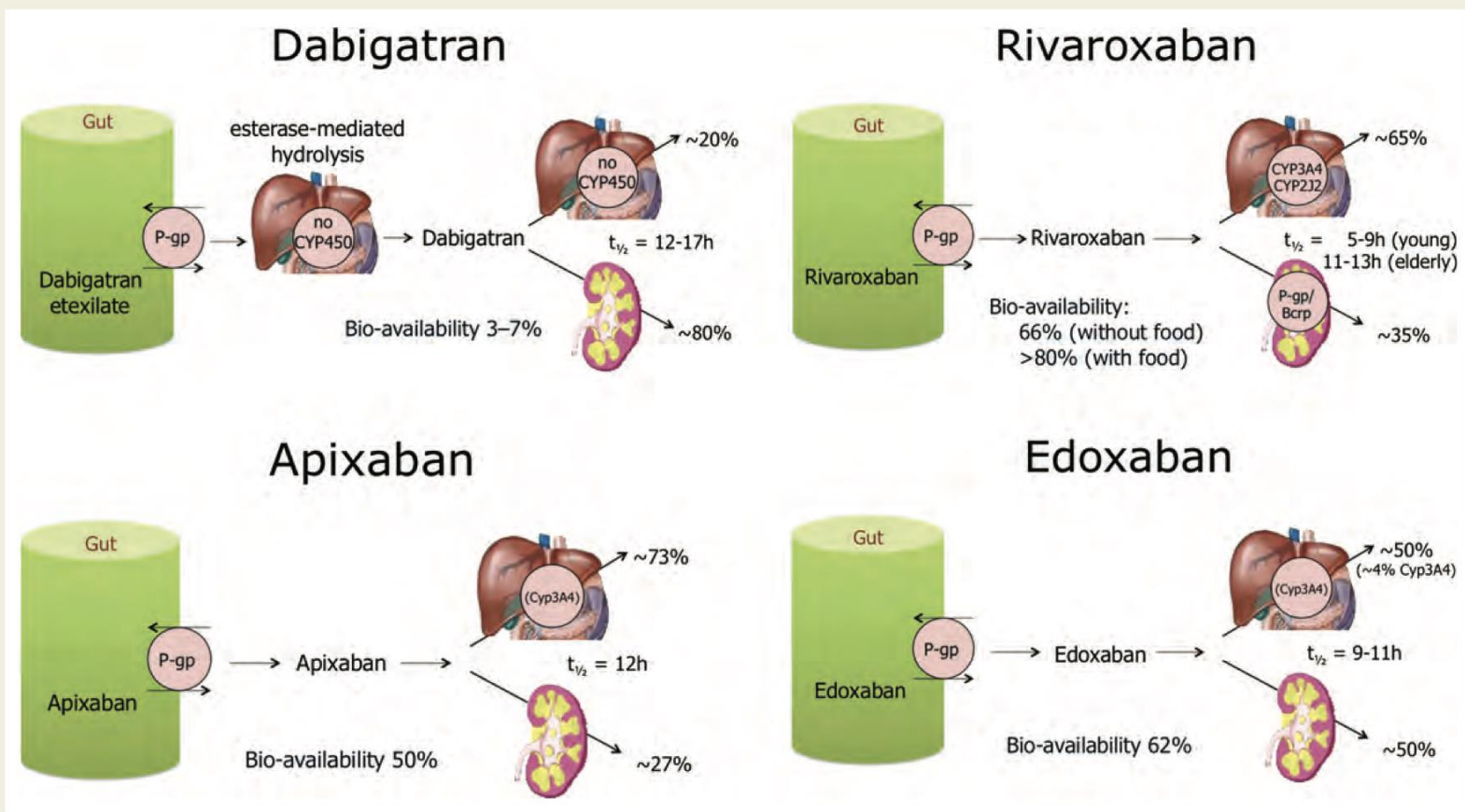


Figure 5 Absorption and metabolism of the different new anticoagulant drugs. There are interaction possibilities at the level of absorption or first transformation, and at the level of metabolisation and excretion. The brackets around (Cyp3A4) in the apixaban graph indicate a minor contribution of this pathway to hepatic clearance, the majority of the drug being excreted as unchanged parent. See also [Table 4](#) for the size of the interactions based on these schemes.

DOACs

Table 6 Estimated drug half-lives and effect on area under the curve NOAC plasma concentrations in different stages of chronic kidney disease compared to healthy controls

	Dabigatran	Apixaban	Edoxaban ^a	Rivaroxaban
CrCl ≥ 60 ml/min CKD Stage I and II	~ 14 h ⁴⁸	No data	~ 8.6 h ⁴⁹	~ 8.5 h ⁵⁰ (+44%)
CrCl 30–60 ml/min CKD Stage III	~ 18 h ⁴⁸	No data	~ 9.4 h ⁴⁹	~ 9 h (+52%)
CrCl 15–30 ml/min CKD Stage IV	~ 28 h ⁴⁸	No data	~ 16.9 h ⁴⁹	~ 9.5 h (+64%)
CrCl ≤ 15 ml/min CKD Stage V	No data	No data	No data	No data

^aNo EMA approval yet. Needs update after finalisation of SmPC.

CKD, chronic kidney disease; CrCl, creatinine clearance.

Hatching, no available data yet.

Table 7 NOACs in renal dysfunction: Approved European labels and dosing in chronic kidney disease

	Dabigatran	Apixaban	Edoxaban ^a	Rivaroxaban
Fraction renally excreted of absorbed dose	80%	27%	50% ⁹	35%
Bio-availability	3–7%	50%	62% ¹⁷	66% without food Almost 100% with food
Fraction renally excreted of administered dose	4%	14%	37% ⁹	33%
Approved for CrCl ≥ ...	≥ 30 ml/min	≥ 15 ml/min	Not available	≥ 15 ml/min
Dosing recommendation	CrCl ≥ 50 ml/min: no adjustment (i.e. 150 mg bid)	Serum creatinine ≥ 1.5 mg/dl: no adjustment (i.e. 5 mg bid)	Not available	CrCl ≥ 50 ml/min: no adjustment (i.e. 20 mg qd)
Dosing if CKD	When CrCl 30–49 ml/min, 150 mg bid is possible (SmPC) but 110 mg bid if 'high risk of bleeding' (SmPC) or 'recommended' (GL update) ² Note: 75 mg bid approved in US only: ^b • if CrCl 15–30 ml/min • if CrCl 30–49 ml/min and other orange factor Table 5 (e.g. verapamil)	CrCl 15–29 ml/min: 2.5 mg bid Serum creatinine ≥ 1.5 mg/dl in combination with age ≥ 80 years or weight ≤ 60 kg ^{SmPC} or with other 'yellow' factor (Table 5): 2.5 mg bid	Not available	15 mg qd when CrCl 15–49 ml/min
Not recommended if	CrCl < 30 ml/min	CrCl < 15 ml/min	Not available	CrCl < 15 ml/min

Orange, reduce dose (from 150 mg BID to 100 mg BID for dabigatran).

Yellow, consider dose reduction if another 'yellow' factor is present (from 20 mg to 15 mg QD for rivaroxaban; from 5 mg BID to 2.5 mg BID for apixaban).

Hatching, no data available yet.

^aNo EMA approval yet. Needs update after finalisation of SmPC.

^bNo EMA indication. FDA recommendation based on pharmacokinetics. Carefully weigh risks and benefits of this approach. Note that 75 mg capsules are not available on the European market for AF indication.

CKD, chronic kidney disease; CrCl, creatinine clearance; bid, twice daily; qd, once daily; SmPC, summary of product characteristics.

DOACs

Table 8 Possible measures to take in case of bleeding

	Direct thrombin inhibitors (dabigatran)	FXa inhibitors (apixaban, edoxaban, rivaroxaban)
Non life-threatening bleeding	Inquire last intake + dosing regimen Estimate normalization of haemostasis: Normal renal function: 12–24 h CrCl 50–80 ml/min: 24–36 h CrCl 30–50 ml/min: 36–48 h CrCl <30 ml/min: ≥48 h Maintain diuresis Local haemostatic measures Fluid replacement (colloids if needed) RBC substitution if necessary Platelet substitution (in case of thrombocytopenia $\leq 60 \times 10^9/L$ or thrombopathy) Fresh frozen plasma as plasma expander (not as reversal agent) Tranexamic acid can be considered as adjuvans Desmopressin can be considered in special cases (coagulopathy or thrombopathy) Consider dialysis (preliminary evidence: -65% after 4 h)⁴⁸ Charcoal haemoperfusion not recommended (no data)	Inquire last intake + dosing regimen Normalization of haemostasis: 12–24 h Local haemostatic measures Fluid replacement (colloids if needed) RBC substitution if necessary Platelet substitution (in case of thrombocytopenia $\leq 60 \times 10^9/L$ or thrombopathy) Fresh frozen plasma as plasma expander (not as reversal agent) Tranexamic acid can be considered as adjuvans Desmopressin can be considered in special cases (coagulopathy or thrombopathy)
Life-threatening bleeding	All of the above Prothrombin complex concentrate (PCC) 25 U/kg (may be repeated once or twice) (but no clinical evidence) Activated PCC 50 IE/kg; max 200 IE/kg/day: no strong data about additional benefit over PCC. Can be considered before PCC if available Activated factor VII (rFVIIa; 90 µg/kg) no data about additional benefit + expensive (only animal evidence)	All of the above Prothrombin complex concentrate (PCC) 25 U/kg (may be repeated once or twice) (but no clinical evidence) Activated PCC 50 IE/kg; max. 200 IE/kg/day: no strong data about additional benefit over PCC. Can be considered before PCC if available Activated factor VII (rFVIIa; 90 µg/kg) no data about additional benefit + expensive (only animal evidence)

RBC, red blood cells; CrCl, creatinine clearance; PCC, prothrombin complex concentrate.

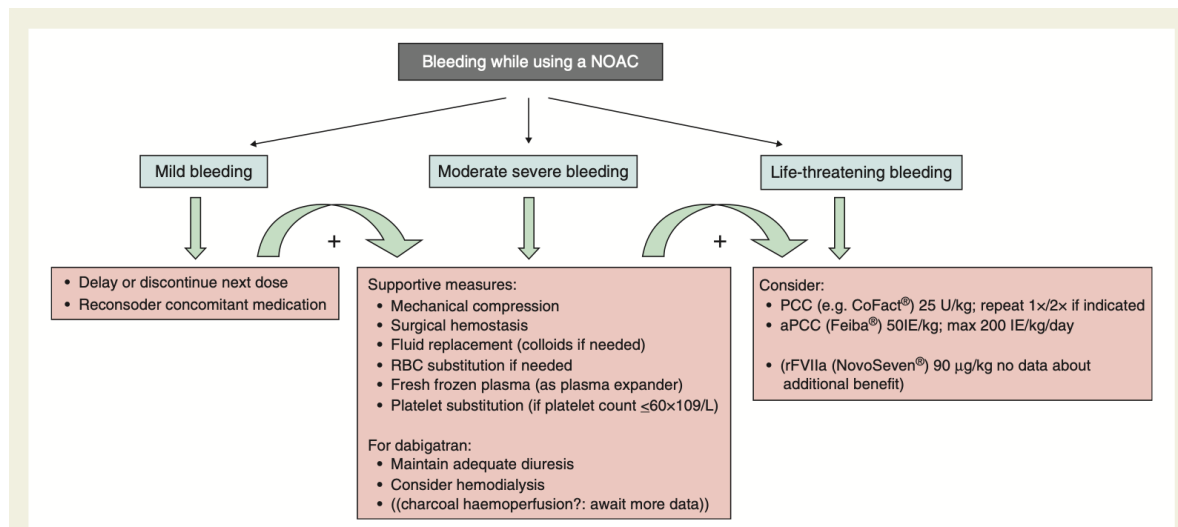


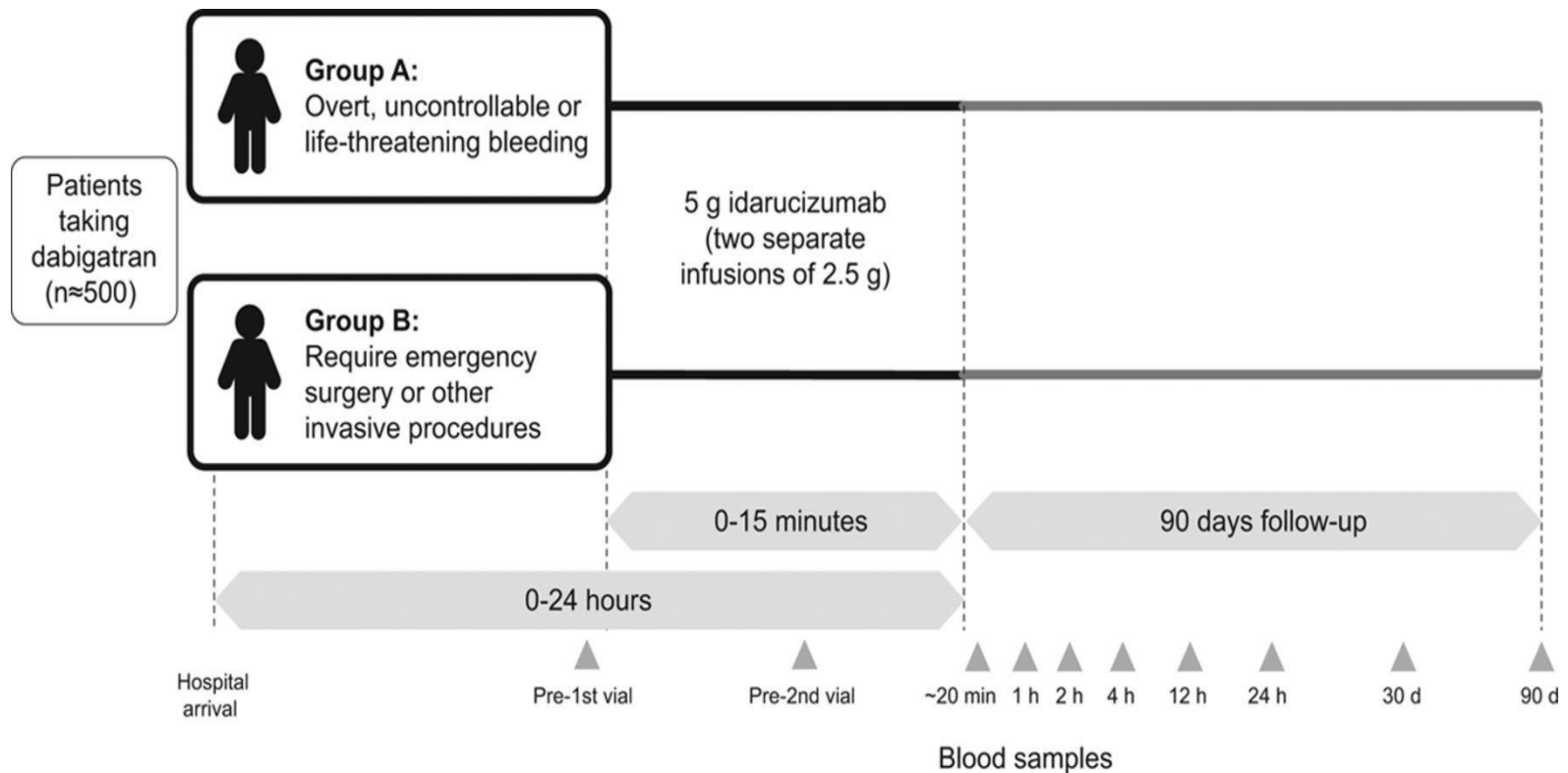
Figure 6 Management of bleeding in patients taking NOACs. Possible therapeutic measures in case of minor or severe bleeding in patients on NOAC therapy. Based on van Ryn et al.¹²

Antidotes for DOACs

TABLE 3. Summary of currently approved antidotes to DOACs and NOACs^{70,71}

Drug	FDA approval date	Target anticoagulants	Mechanism of action	Route of administration	Typical available dose
Idarucizumab	October 16, 2015	-Dabigatran	-humanized monoclonal antibody that binds dabigatran and its active metabolites.	IV	-5 g -2.5 g
Andexanet alfa	May 3, 2018	-Rivaroxaban -Apixaban -Edoxaban -LMWH -Fondaparinux	-Binds and sequesters the Factor Xa -inhibit activity of Tissue Factor Pathway Inhibitor (TFPI) -increased tissue factor driven thrombin generation.	IV	-100 mg -200 mg
Aripazine Ciraparantag PER977	April 2015 (FDA fast-track designation)	-Rivaroxaban -Dabigatran -Apixaban -Edoxaban -UFH -LMWH –Fondaparinux		IV	100-400 mg
4-factor PCC	April 30, 2013	-Dabigatran -Rivaroxaban -Apixaban -Edoxaban-warfarin	-supplies additional clotting factors.	IV	500-1000 units 20-40 ml

Abbreviation: IV, intravenous, UFH, unfractionated heparin, LMWH, low molecular weight heparin, PCC, prothrombin complex concentrate



$$\% \text{ reversal} = \frac{[\text{pre-dose test result (seconds)} - \text{minimum post-dose test results (seconds)}]}{[\text{pre-dose test result (seconds)} - \text{upper limit of normal}]} \times 100$$



Idarucizumab

- Idarucizumab (Praxbind) is a humanized monoclonal antibody fragment against dabigatran which FDA approved in 2015 to reverse dabigatran-associated bleeding in emergency surgeries and urgent procedures and life-threatening or uncontrolled bleeding. It is administered intravenously in 2 separate 2.5-gram dosages 15 minutes apart. Time to onset of action is within minutes, and hemostasis is achieved in a median of 11.4 hours in cases of life threatening bleeding. The duration of action in the body is at least 24 hours. Thirty-two percent of the molecule is excreted from the body within the first 6 hours of administration and less than 1% of it in the following 18 hours. The rest of the drug is assumed to be eliminated via protein catabolism, mainly in the kidneys. Possible adverse events related to Idarucizumab include thromboembolic risks,
- TaggedEndCurr Probl Cardiol, February 2023 11
- hypersensitivity reactions, and severe adverse reactions in patients with hereditary fructose intolerance due to sorbitol excipient. Anticoagulant therapy should be re-initiated to reduce its thrombotic risk as soon as the patient's medical condition stabilizes.[46,47,48](#) A phase 3 case series study showed that idarucizumab completely reverses dabigatran in >98% of patients regardless of renal function. Although re-elevation of dabigatran levels within 12-24 hours is commonly performed, the time to bleeding cessation and the extent of hemostasis during procedures are similar.

Andexanet alfa

- Andexanet alfa is a recombinant inactive factor Xa derivative used to reverse the effect of direct factor Xa inhibitors and indirect factor Xa inhibitors (LMWH, enoxaparin, fondaparinux). FDA indicated it in 2018 for patients treated with rivaroxaban and apixaban when reversal of anticoagulation is needed due to life-threatening or uncontrolled bleeding. It is administered intravenously, and the average elimination half-life is 5-7 hours. The most frequently encountered adverse effects are pneumonia and urinary tract infections.⁵¹ Life-threatening adverse effects of the drug include:
 - Arterial and venous thromboembolic events. Tagged End Ischemic events (eg, MI, stroke).
Cardiac arrest.
Sudden deaths.
- The drug cannot be detected in urine, and the excretion route of the medicine is unknown currently; however, studies to enlighten pharmacokinetic/dynamic properties, safety, and efficacy of the molecule are being carried on.⁵² The 18-month single-center observational study of real-world use of andexanet alfa in patients with factor Xa inhibitor-associated intracranial hemorrhage concluded in an excellent hemostatic efficacy rate, lower inpatient mortality, and 30-day readmission due to thromboembolism; findings similar to those reported in ANNEXA-4 study. Sub-group analysis of 1 phase 3 study showed that andexanet was effective and safe in lowering the anticoagulant activity of FXa for up to 12 hours after administration in patients with intracranial bleeding.⁵¹ A randomized, controlled phase 4 study with the largest participants to date is being conducted to evaluate and compare the efficacy and safety of andexanet alfa versus the usual standard of care in patients with intracranial hemorrhage anticoagulated with DOACs.⁵³ Annexa-S is a multicentre, prospective, open-label phase 2 study being conducted to evaluate the efficacy and safety of andexanet alfa in patients requiring urgent surgery who have been anticoagulated with FXa (activated factor X) inhibitors.⁵² ASTRO-DE is a multicenter, observational study that is being conducted
- to provide real-life data about patients with intracranial hemorrhage under adequate anticoagulation with rivaroxaban and apixaban, who are treated with andexanet alfa.

DOACs and Preoperative Management

Table 9 Last intake of drug before elective surgical intervention

	Dabigatran		Apixaban		Edoxaban ^a		Rivaroxaban	
No important bleeding risk and/or adequate local haemostasis possible: perform at trough level (i.e. ≥ 12 h or 24 h after last intake)								
	Low risk	High risk	Low risk	High risk	Low risk	High risk	Low risk	High risk
CrCl ≥ 80 ml/min	≥ 24 h	≥ 48 h	≥ 24 h	≥ 48 h	No data	No data	≥ 24 h	≥ 48 h
CrCl 50–80 ml/min	≥ 36 h	≥ 72 h	≥ 24 h	≥ 48 h	No data	No data	≥ 24 h	≥ 48 h
CrCl 30–50 ml/min ^b	≥ 48 h	≥ 96 h	≥ 24 h	≥ 48 h	No data	No data	≥ 24 h	≥ 48 h
CrCl 15–30 ml/min ^b	Not indicated	Not indicated	≥ 36 h	≥ 48 h	No data	No data	≥ 36 h	≥ 48 h
CrCl < 15 ml/min	No official indication for use							

Bold values deviate from the common stopping rule of ≥ 24 h low risk, ≥ 48 h high risk.

^aNo EMA approval yet. Needs update after finalisation of SmPC.

^bMany of these patients may be on the lower dose of dabigatran (i.e. 110 mg BID) or apixaban (i.e. 2.5 mg BID), or have to be on the lower dose of rivaroxaban (15 mg QD).

Low risk = surgery with low risk of bleeding; high risk = surgery with high risk of bleeding. See also Table 10.

CrCl, creatinine clearance.

Direct Oral Anticoagulants

Table 2. DOAC Use in Valvular Heart Disease

	Dabigatran	Rivaroxaban	Apixaban	Edoxaban	Betrixaban
Mechanical prosthetic valve	Contraindicated				NA*
Moderate to severe mitral stenosis	Contraindicated				NA*
Bioprosthetic valve [†]	Acceptable				NA*

NA indicates not applicable.

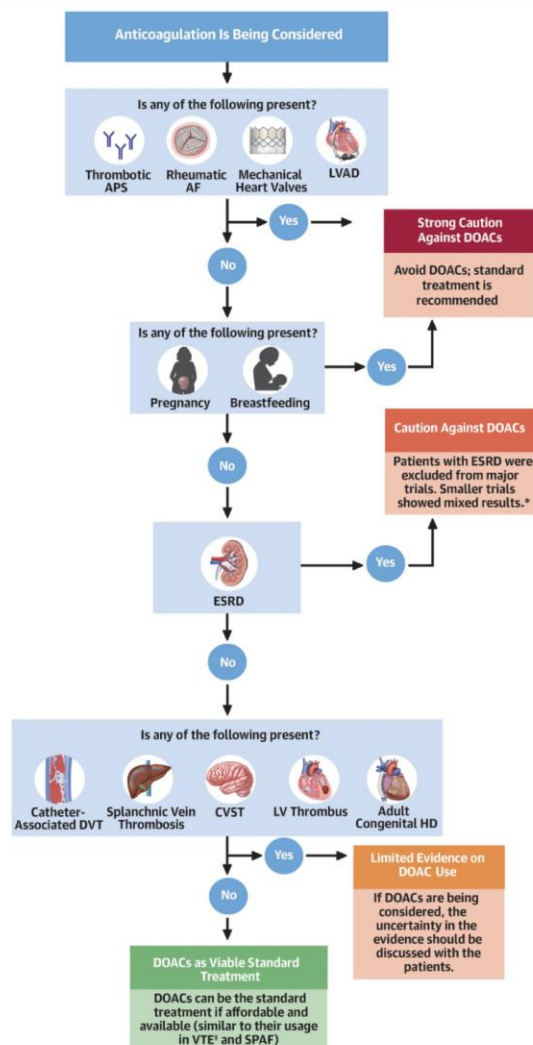
*Betrixaban is used only as a prophylactic agent; it is not indicated for treatment.

[†]Not advised for rheumatic mitral stenosis.

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Conditions Not Suitable for DOAC

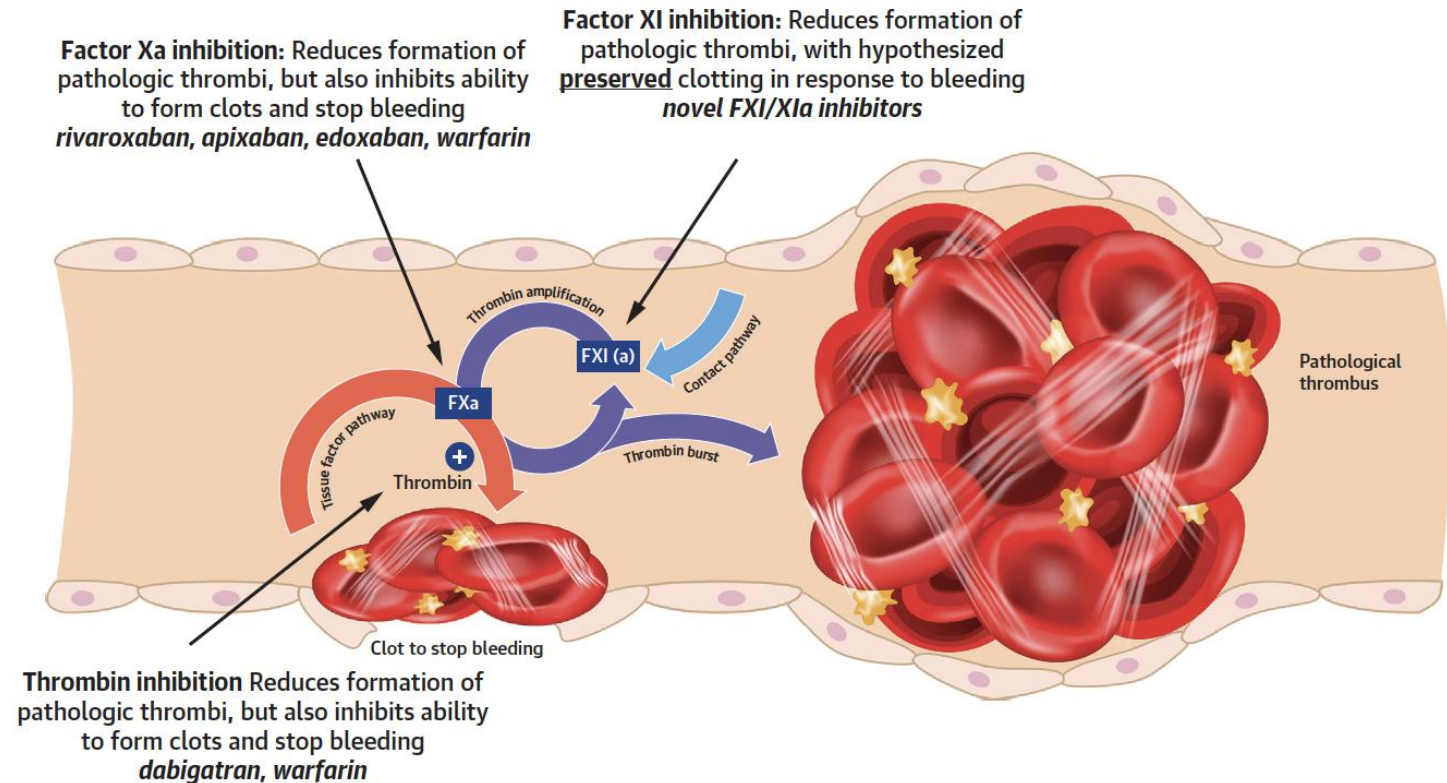
CENTRAL ILLUSTRATION: Important Considerations for Direct Oral Anticoagulant Use



- Mechanical heart valves
- Thrombotic antiphospholipid syndrome
- AF associated with chronic rheumatic heart disease

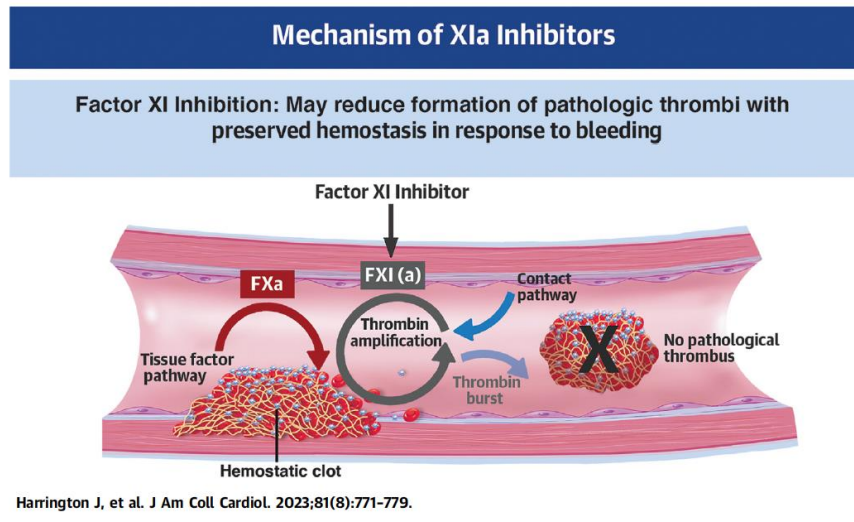
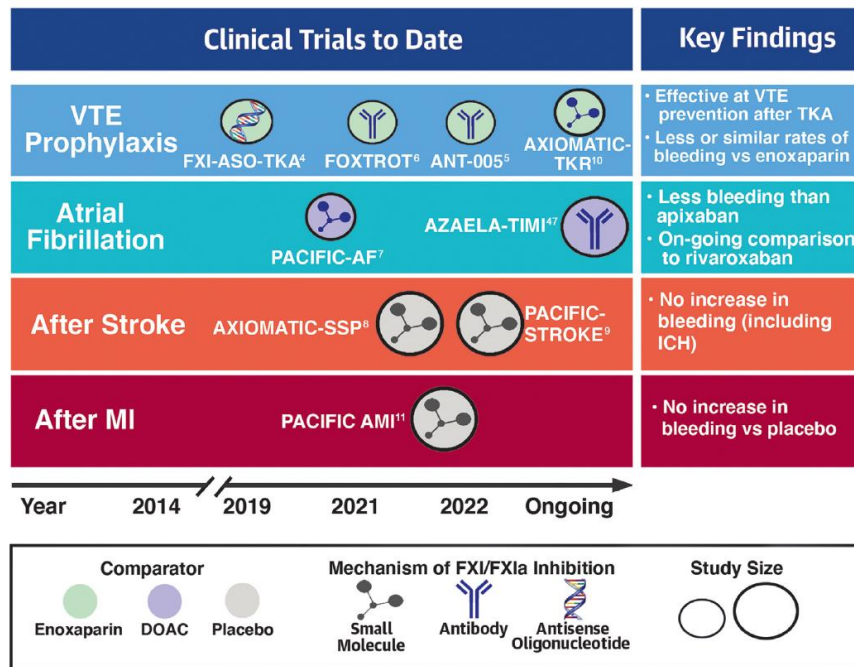
Factor XI Inhibitors

FIGURE 1 The Coagulation Cascade and Mechanisms of Different Anticoagulants



Direct oral anticoagulants and warfarin all act on the common pathway, inhibiting both hemostasis and pathologic thrombus formation. By inhibiting FXI, it may be possible to reduce the risk of thrombi while preserving the functionality of the clotting cascade in response to bleeding.

CENTRAL ILLUSTRATION Clinical Data and Proposed Mechanism of Factor XI/Factor XIa Inhibition



Proposed mechanism of factor XI (FXI) inhibitors and key findings and clinical trials of FXI inhibitors to date, including trial size, mechanism of drug, and control arm. ICH = intracranial hemorrhage; MI = myocardial infarction; TKA = total knee arthroscopy; VTE = venous thromboembolism.

ORIGINAL ARTICLE

Asundexian versus Apixaban in Patients with Atrial Fibrillation

J.P. Piccini, M.R. Patel, J. Steffel, K. Ferdinand, I.C. Van Gelder, A.M. Russo, C.-S. Ma, S.G. Goodman, J. Oldgren, C. Hammett, R.D. Lopes, M. Akao, R. De Caterina, P. Kirchhof, D.A. Gorog, M. Hemels, M. Rienstra, W.S. Jones, J. Harrington, G.Y.H. Lip, S.J. Ellis, F.W. Rockhold, C. Neumann, J.H. Alexander, T. Viethen, J. Hung, R. Coppolecchia, H. Mundl, and V. Caso, for the OCEANIC-AF Steering Committee and Investigators*

ABSTRACT

BACKGROUND

Stroke prevention with direct-acting oral anticoagulant agents in patients with atrial fibrillation confers a risk of bleeding and limits their use. Asundexian, an activated factor XI (XIIa) inhibitor, is an oral anticoagulant that may prevent strokes with less bleeding.

METHODS

In a phase 3, international, double-blind trial, we randomly assigned high-risk patients with atrial fibrillation in a 1:1 ratio to receive asundexian at a dose of 50 mg once daily or standard-dose apixaban. The primary efficacy objective was to determine whether asundexian is at least noninferior to apixaban for the prevention of stroke or systemic embolism. The primary safety objective was to determine whether asundexian is superior to apixaban with respect to major bleeding events.

RESULTS

A total of 14,810 randomly assigned patients were included in the intention-to-treat population. The mean ($\pm$ SD) age of the patients was 73.9 $\pm$ 7.7 years, 35.2% were women, 18.6% had chronic kidney disease, 18.2% had a previous stroke or transient ischemic attack, 16.8% had received oral anticoagulants for no more than 6 weeks, and the mean CHA₂DS₂-VASc score (range, 0 to 9, with higher scores indicating a greater risk of stroke) was 4.3 $\pm$ 1.3. The trial was stopped prematurely at the recommendation of the independent data monitoring committee. Stroke or systemic embolism occurred in 98 patients (1.3%) assigned to receive asundexian and in 26 (0.4%) assigned to receive apixaban (hazard ratio, 3.79; 95% confidence interval [CI], 2.46 to 5.83). Major bleeding occurred in 17 patients (0.2%) who received asundexian and in 53 (0.7%) who received apixaban (hazard ratio, 0.32; 95% CI, 0.18 to 0.55). The incidence of any adverse event appeared to be similar in the two groups.

CONCLUSIONS

Among patients with atrial fibrillation at risk for stroke, treatment with asundexian at a dose of 50 mg once daily was associated with a higher incidence of stroke or systemic embolism than treatment with apixaban in the period before the trial was stopped prematurely. There were fewer major bleeding events with asundexian than with apixaban during this time. (Funded by Bayer; OCEANIC-AF ClinicalTrials.gov number, NCT05643573; EudraCT number, 2022-000758-28.)

The authors' full names, academic degrees, and affiliations are listed in the Appendix. Dr. Piccini can be contacted at jonathan.piccini@duke.edu or at the Duke Clinical Research Institute, Duke University Medical Center, 300 W. Morgan St., Durham, NC 27701.

*A complete listing of the OCEANIC-AF steering committee and investigators is provided in the Supplementary Appendix, available at NEJM.org.

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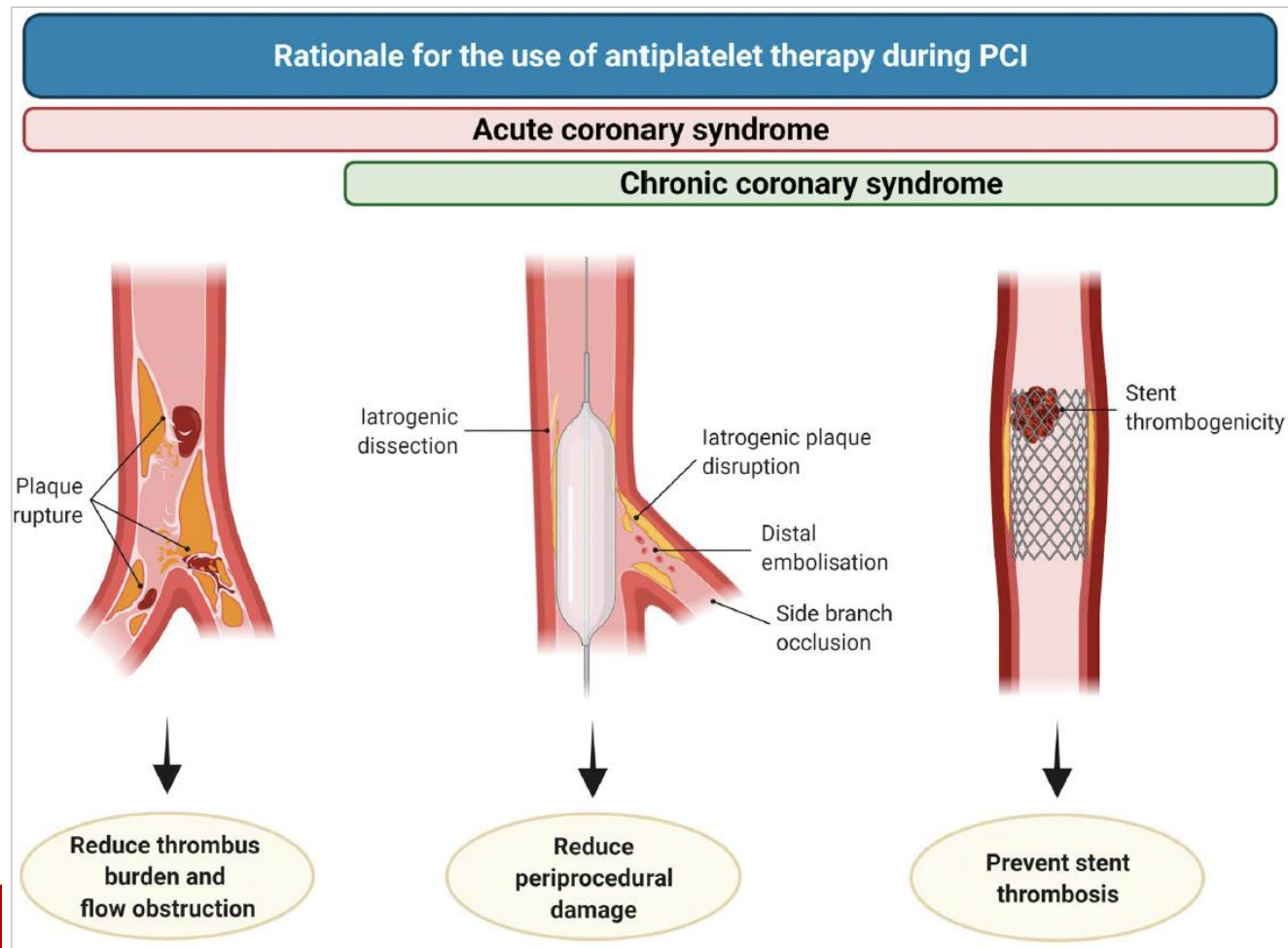


Figure 2. Rationale for the use of antiplatelet therapy during PCI. In patients undergoing PCI in the setting of chronic coronary syndrome (CCS), antiplatelet therapy reduces the occurrence of intraprocedural or very early stent thrombosis (right), periprocedural damage caused by iatrogenic dissections, plaque disruption, distal embolisation or side branch occlusion (middle). During acute coronary syndrome, antiplatelet therapy also plays a role in reducing thrombus burden and flow obstruction (left). PCI: percutaneous coronary intervention.



Antiplatelet therapy after percutaneous coronary intervention

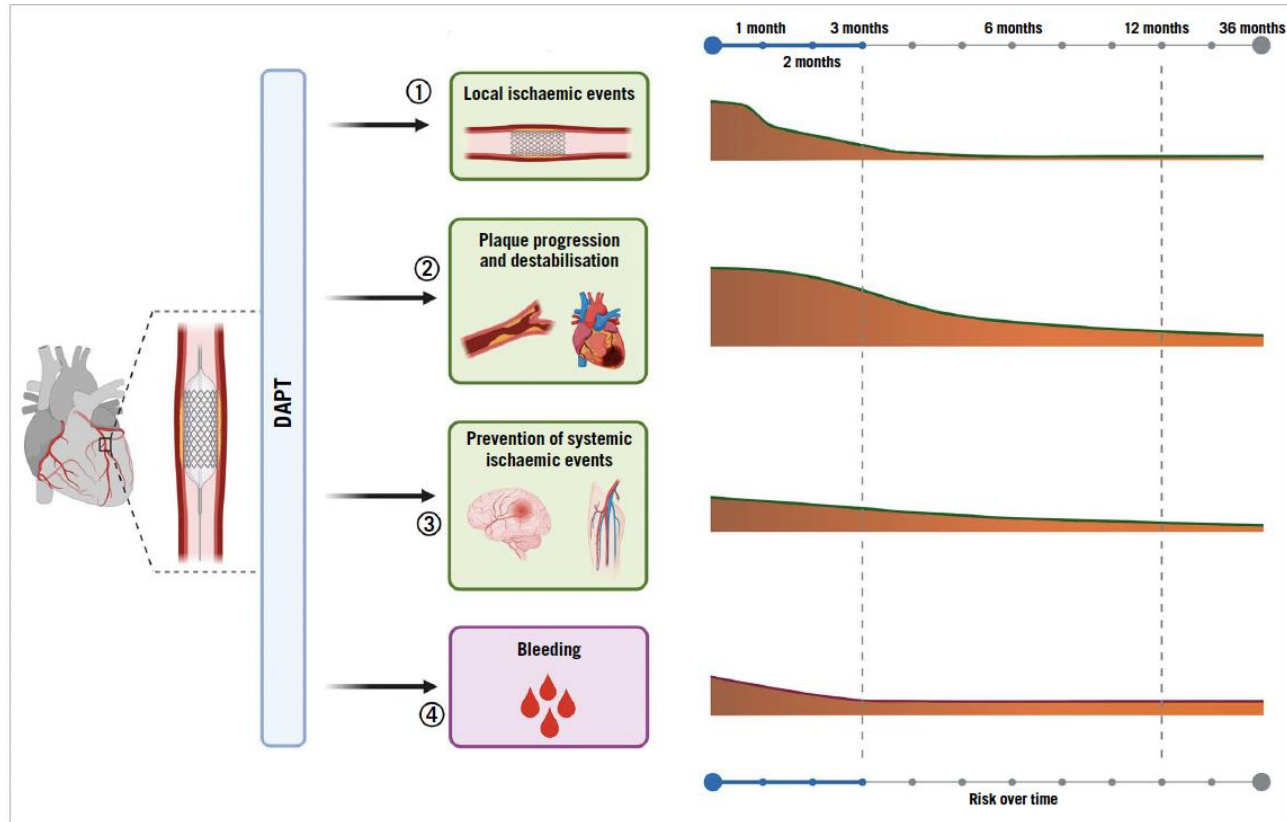


Figure 3. Time course of benefit and risk of antiplatelet therapy after PCI. Antiplatelet agents administered after PCI may 1) reduce the incidence of stent-related ischaemic events such as stent thrombosis or target vessel revascularisation; 2) reduce the incidence of cardiovascular ischaemic recurrence and their consequences, such as myocardial infarction and cardiovascular death; 3) prevent cerebrovascular events in other areas affected by atherosclerotic disease, such as peripheral and carotid arteries; 4) be inevitably associated with increased risk of bleeding. Importantly, the potential benefit of antiplatelet agents varies over time, with the greatest benefit in terms of less ischaemic events maximised during the first months after PCI and decreasing over time, while bleeding events remain stable over time. DAPT: dual antiplatelet therapy; PCI: percutaneous coronary intervention.



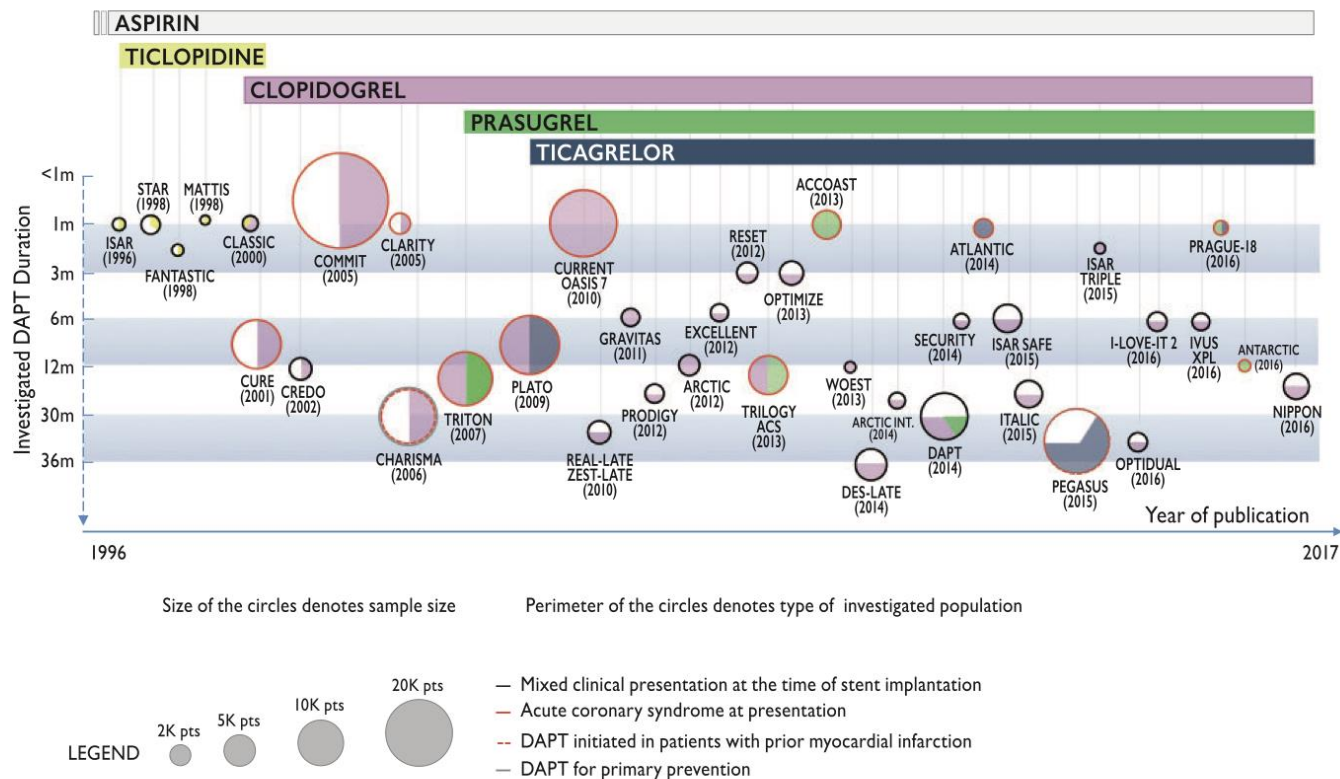
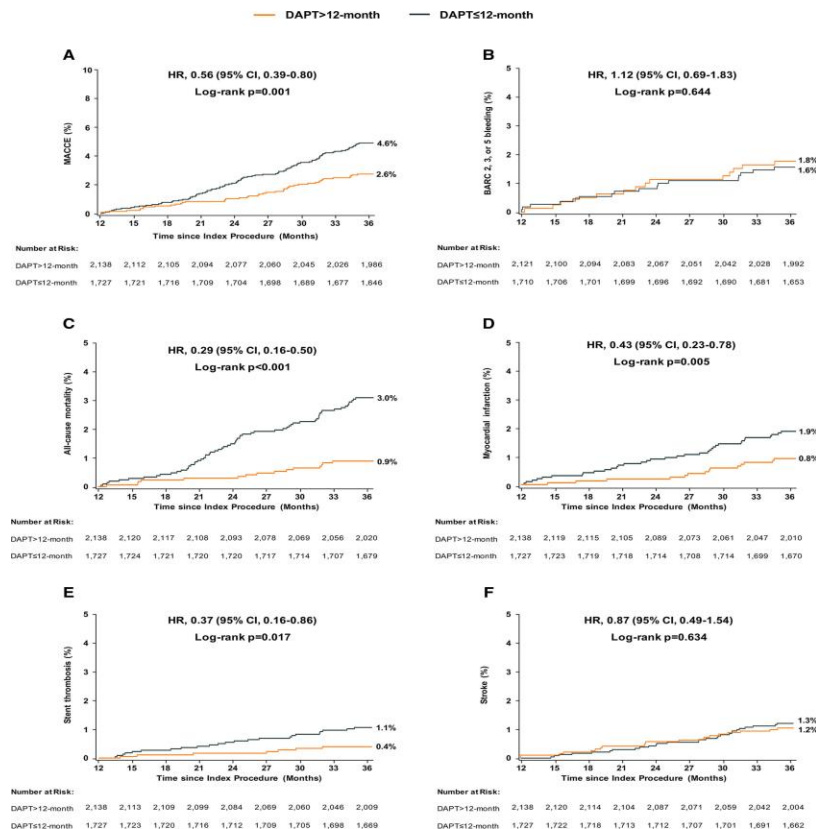


Figure 1 History of dual antiplatelet therapy (DAPT) in patients with coronary artery disease. The size of the circles denotes sample size. The colours of perimeters identify the type of included patient populations within each study. The colours within each circle identify the antiplatelet agent(s) investigated. Head-to-head studies comparing similar durations of two different antiplatelet strategies are shown with a vertical line, whereas those investigating different treatment durations are shown with a horizontal line. Studies investigating different treatment strategies or regimens and not treatment durations or type (e.g. pre-treatment in ACCOAST, tailored therapy in GRAVITAS, double dose of clopidogrel in CURRENT OASIS 7, etc.) are represented with a single colour indicating the P2Y₁₂ inhibitor, which was tested on top of aspirin. pts = patients.

Extended DAPT Rx in Patients with LM Stenting



Hao-Yu Wang. Circulation: Cardiovascular Interventions. New Insights Into Long- Versus Short-Term Dual Antiplatelet Therapy Duration in Patients After Stenting for Left Main Coronary Artery Disease: Findings From a Prospective Observational Study, Volume: 15, Issue: 6, Pages: e011536, DOI: (10.1161/CIRCINTERVENTIONS.121.011536)



	Before PCI	During PCI	After PCI
Stable 	Aspirin Clopidogrel (high probability of PCI) I A IIb C	Clopidogrel Prasugrel or ticagrelor over Clopidogrel (high ischaemic risk) I A IIb C	DAPT for 6 months Extended DAPT (high ischaemic and low bleeding risk) 3-month DAPT → aspirin (high bleeding risk) 1-month DAPT → aspirin (high bleeding risk) I A IIa A IIa A IIb C
NSTE-ACS 	Aspirin Routine P2Y₁₂ inhibitor I A III A	Prasugrel Ticagrelor Clopidogrel Prasugrel over ticagrelor I A I A I C IIa B	DAPT for 12 months Extended DAPT or DPI (high ischaemic risk) 1-month DAPT → clopidogrel (very high bleeding risk) 3-month DAPT → aspirin (high bleeding risk) 3-month DAPT with ticagrelor → ticagrelor Extended DAPT or DPI (moderate ischaemic risk) Guided de-escalation I A IIa A IIa B IIa B IIa B IIb A IIb A
STE-ACS 	Aspirin Potent P2Y₁₂ inhibitor I B I A		DAPT for 12 months 6-month DAPT → aspirin (high bleeding risk) Extended DAPT or DPI (high ischaemic/low bleeding risk) Guided de-escalation I A IIa B IIb B IIb A



Dual antiplatelet therapy duration in patients with acute coronary syndrome treated with percutaneous coronary intervention

Recommendations	Class ^a	Level ^b
In patients with ACS treated with coronary stent implantation, DAPT with a P2Y ₁₂ inhibitor on top of aspirin is recommended for 12 months unless there are contraindications such as excessive risk of bleeding (e.g. PRECISE-DAPT ≥ 25). ^{20,23,40}	I	A
In patients with ACS and stent implantation who are at high risk of bleeding (e.g. PRECISE-DAPT ≥ 25), discontinuation of P2Y ₁₂ inhibitor therapy after 6 months should be considered. ^{13,18,143}	IIa	B
In patients with ACS treated with bioresorbable vascular scaffolds, DAPT for at least 12 months should be considered.	IIa	C
In patients with ACS who have tolerated DAPT without a bleeding complication, continuation of DAPT for longer than 12 months may be considered. ^{26,139}	IIb	A
In patients with MI and high ischaemic risk ^c who have tolerated DAPT without a bleeding complication, ticagrelor 60 mg <i>b.i.d.</i> for longer than 12 months on top of aspirin may be preferred over clopidogrel or prasugrel. ^{29,115,142}	IIb	B

ACS = acute coronary syndrome; *b.i.d.* = *bis in die*; DAPT = dual antiplatelet therapy. MI = myocardial infarction; PRECISE-DAPT = PREdicting bleeding Complications In patients undergoing Stent implantation and subSequent Dual Anti Platelet Therapy.

^aClass of recommendation.

^bLevel of evidence.

^cDefined as ≥ 50 years of age, and one or more of the following additional high-risk features: age of 65 years or older, diabetes mellitus requiring medication, a second prior spontaneous myocardial infarction, multivessel coronary artery disease, or chronic renal dysfunction, defined as an estimated creatinine clearance < 60 mL/min.

These recommendations refer to stents that are supported by large-scale randomized trials with clinical endpoint evaluation leading to unconditional CE mark, as detailed in Byrne *et al.*¹³⁴

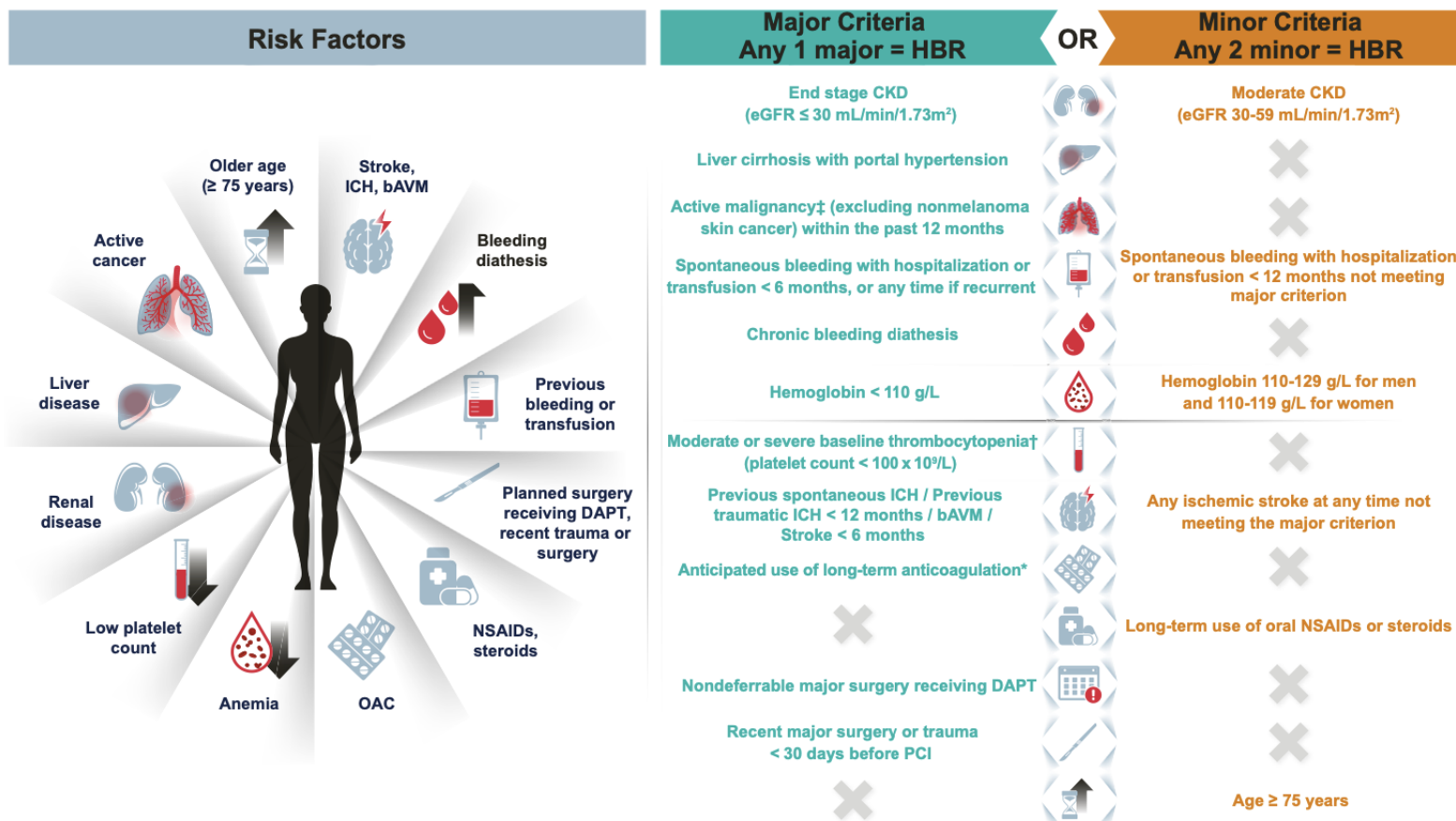
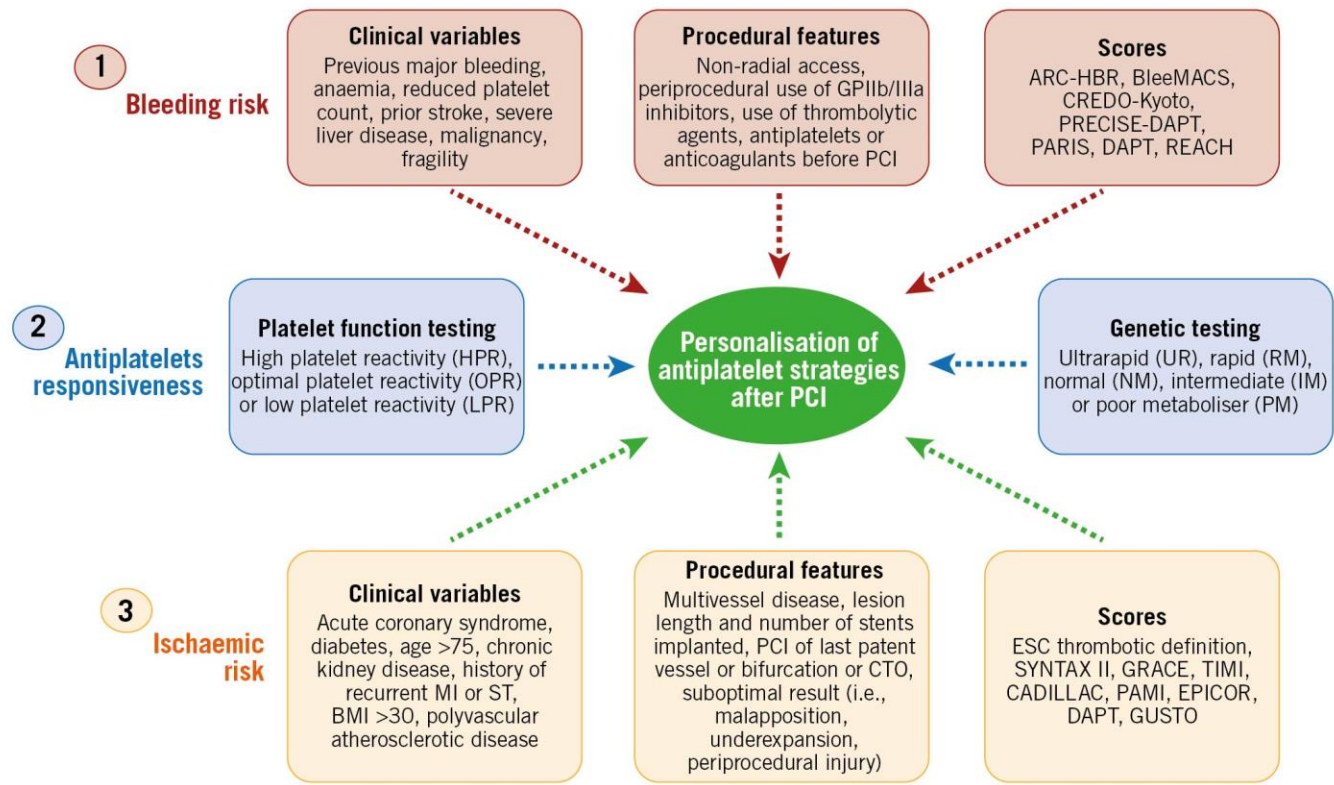


Figure 2. Estimating bleeding risk for antiplatelet decision-making after percutaneous coronary intervention (PCI). This illustration provides criteria for high bleeding risk (HBR) as defined by the Bleeding Academic Research Consortium (1 major or 2 minor criteria), when short dual antiplatelet therapy (DAPT) should be considered. bAVM, brain arteriovenous malformation; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; ICH, intracranial hemorrhage; NSAID, nonsteroidal anti-inflammatory drug; OAC, oral anticoagulation. * Excludes vascular protection doses. † Baseline thrombocytopenia is defined before PCI. ‡ Active malignancy is defined as a diagnosis within 12 months and/or ongoing treatment.



2020 ACC Expert Consensus Decision Pathway for Anticoagulant and Antiplatelet Therapy

1. Approximately 10% of patients with recent PCI have concomitant AF. Others may have concomitant VTE. Choosing the optimal antithrombotic regimen can be a challenge.
2. Clinical pathways are suggested for four potential clinical situations: (1) prior AF on anticoagulation and the need for PCI; (2) new-onset AF requiring anticoagulation in a patient already on antiplatelet therapy for CAD; (3) prior VTE on anticoagulation and the need for PCI; and (4) new or recurrent VTE requiring anticoagulation in a patient already on antiplatelet therapy for CAD.
3. In general, the use of “triple therapy” is not recommended for most patients due to an increased risk of bleeding. A short duration (e.g., no more than 30 days) is recommended. When combined with an anticoagulant, clopidogrel is the recommended antiplatelet agent for most patients. If aspirin is being used, it should be limited to <100 mg daily dosing.
4. For patients taking ≥ 2 antithrombotic agents, starting or continuing a proton pump inhibitor and avoiding other anti-inflammatory medications should be employed to reduce gastrointestinal bleeding risk.
5. For patients with AF on anticoagulation who need a PCI, use of a direct oral anticoagulant (DOAC) is preferred over VKA when appropriate. Oral anticoagulation plus P2Y₁₂ antiplatelet combination is recommended for the first 6-12 months (potentially switching P2Y₁₂ to aspirin for months 6-12 if PCI for stable ischemic heart disease), followed by anticoagulation monotherapy after 12 months.
6. For patients on antiplatelet therapy who develop new AF, management depends on the indication for antiplatelet therapy. For primary prevention, switch to anticoagulation monotherapy is recommended. For PCI with stable ischemic heart disease or acute coronary syndrome, use of oral anticoagulant plus a P2Y₁₂ inhibitor for no more than 12 months is recommended, followed by oral anticoagulation alone. For patients with cerebrovascular disease without carotid stenting, oral anticoagulation monotherapy is recommended. For patients with carotid stenting or peripheral artery disease, a short course of anticoagulation plus P2Y₁₂ inhibitor may be recommended, followed by oral anticoagulation alone.

2020 ACC Expert Consensus Decision Pathway for Anticoagulant and Antiplatelet Therapy

7. Management of patients with prior VTE depends on the clinical situation during which the VTE event occurred. If the VTE was associated with strongly provoking, reversible risk factors, then delaying a PCI may be beneficial so that anticoagulation therapy can be discontinued. If long-term/indefinite anticoagulation is required, then use of standard treatment doses of anticoagulation plus P2Y₁₂ inhibitor antiplatelet therapy are recommended following PCI. If the dose of anticoagulation is reduced (e.g., “half-dose DOAC” for VTE secondary prevention), then continued use of a single antiplatelet medication (e.g., aspirin) is indicated long-term
8. For patients on antiplatelet therapy who develop a new VTE event, use of anticoagulation plus single antiplatelet medication is generally recommended. Patients using antiplatelet therapy for primary cardiovascular disease prevention or >12 months from the most recent PCI or acute coronary syndrome can be treated with anticoagulation monotherapy.
9. For patients taking DOAC medications who require PCI, most DOACs can be held for no more than 36-48 hours prior to the procedure. Holding a DOAC for longer periods of time may be required for patients with moderate-severe renal dysfunction, especially if using dabigatran.
10. When used in combination with antiplatelet medications, dosing of DOAC medications usually follows the Food and Drug Administration guidance for stroke prevention in AF or treatment of VTE. However, rivaroxaban may be administered at 15 mg daily (reduce to 10 mg daily for creatinine clearance <50 ml/min) when combined with P2Y₁₂ inhibitors.

Antithrombotic Rx Strategies in AF and PCI/ ACS

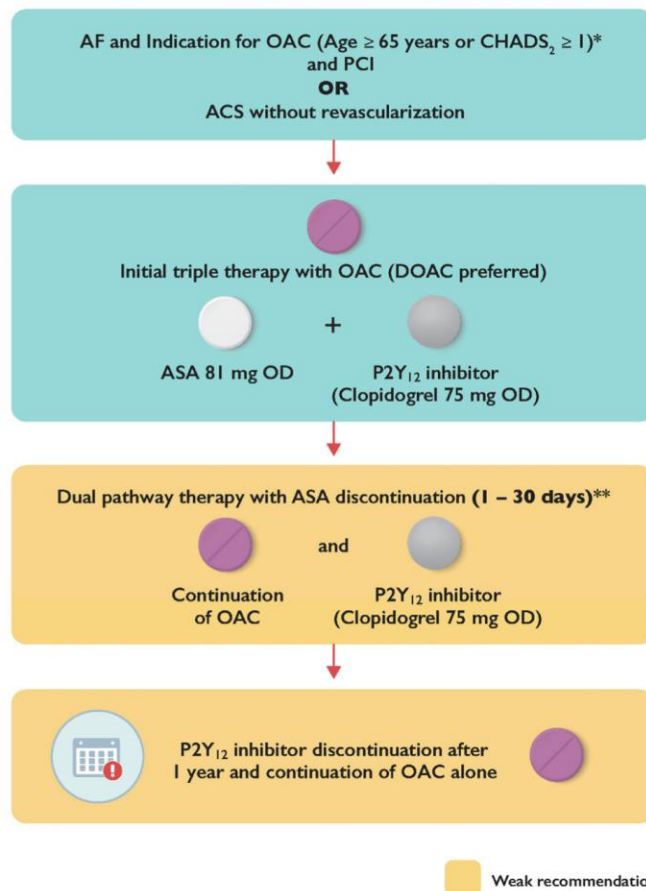


Figure 9. Antiplatelet strategies in patients with concomitant atrial fibrillation (AF) and percutaneous coronary intervention (PCI) and/or acute coronary syndrome (ACS). ASA, acetylsalicylic acid; CHADS₂, Congestive Heart Failure, Hypertension, Age ≥ 75 , Diabetes, and Prior Stroke/Transient Ischemic Attack [doubled]; DOAC, direct oral anticoagulation; OAC, oral anticoagulation; OD, once daily. * Patients who do not require OAC for their AF as per the Canadian guidelines for the management of AF would receive dual antiplatelet therapy.⁸³ OAC regimens evaluated in this context include apixaban 5 mg twice daily (BID); 2.5 mg BID if patients meet 2 or more of the following dosage reduction criteria: older than 80 years of age, < 60 kg in weight, or creatinine > 133 μ M/L, dabigatran 110 mg, or 150 mg BID, edoxaban 60 mg daily (30 mg daily if patients with any of the following characteristics: moderate or severe renal impairment [creatinine clearance 15-50 mL/min], < 60 kg in weight, or concurrent use of specific potent P-glycoprotein inhibitors), rivaroxaban 15 mg daily (10 mg in patients with renal dysfunction), and warfarin. If warfarin is to be used, recommended international normalized ratio target is 2.0-2.5. All patients should receive a loading dose of ASA 160 mg at the time of PCI (if previously ASA-naïve). ** ASA duration before switching to dual pathway was an average of 1.6-6.6 days after PCI in the large-scale trials.

Hospital Authority Indications

Clopidogrel (Plavix)

- Post coronary / vascular intervention (4-week prescription for post-Bare Metal Stenting)
- Post coronary / vascular intervention (1-year prescription for post-Drug Eluting Stenting)
- High risk acute coronary syndrome (up to 1year prescription for patients with acute myocardial infarction or acute coronary syndrome)
- Hypersensitivity to aspirin.
- Secondary prevention for high risk stroke/TIA (3 -13 weeks)

Hospital Authority Indications

Prasugrel (Effient)

- Immediate PCI for STEMI (4-wk prescription for post-Bare Metal Stenting)
- Immediate PCI for STEMI (1-yr prescription for post-Drug Eluting Stenting)
- Urgent PCI for high risk acute coronary syndrome
- Stent thrombosis occurred during clopidogrel (4-wk prescription for post-Bare Metal Stenting)
- Stent thrombosis occurred during clopidogrel (1-yr prescription for post-Drug Eluting Stenting)
- Hypersensitivity to clopidogrel (4-wk prescription for post-Bare Metal Stenting)
- Hypersensitivity to clopidogrel (1-yr prescription for post-Drug Eluting Stenting)

Hospital Authority Indications Ticagrelor (Brilinta)

- All NSTEMI patients waiting for or treated with PCI
- Immediate PCI for STEMI (4-wk prescription for post-Bare Metal Stenting)
- Immediate PCI for STEMI (1-yr prescription for post-Drug Eluting Stenting)
- Urgent PCI for high risk acute coronary syndrome
- Stent thrombosis occurred during clopidogrel (4-wk prescription for post-Bare Metal Stenting)
- Stent thrombosis occurred during clopidogrel (1-yr prescription for post-Drug Eluting Stenting)
- Hypersensitivity to clopidogrel (4-wk prescription for post-Bare Metal Stenting)
- Hypersensitivity to clopidogrel (1-yr prescription for post-Drug Eluting Stenting)
- Given with aspirin up to 3 years (continue therapy after 1st year of Dual Antiplatelet Therapy) for high risk MI patient without bleeding complication

Hospital Authority Indications Eptifibatide (Integrilin)

- In primary PCI for STEMI and PCI for high

Drug Interactions: PPI and Clopidogrel

- Clopidogrel requires metabolic transformation in the liver by CYP isoenzymes (mainly CYP2C19) to elicit its antiplatelet effect.
- PPIs are also metabolized by CYP enzymes, leading to a potential inhibition of CYP2C19 (mainly omeprazole and esomeprazole) translating into reduced metabolic activation of clopidogrel when taken together.
- Pharmacodynamic studies demonstrated the reduction of clopidogrel-induced antiplatelet effects when a PPI, mainly omeprazole, was concomitantly administered.
- Based on drug–drug interaction studies, omeprazole and esomeprazole would appear to have the highest propensity for clinically relevant interactions, lansoprazole an intermediate probability, while pantoprazole and rabeprazole have the lowest.
- However, importantly, no interaction between concomitant use of PPI and ticagrelor

JAMA | Original Investigation

Serious Bleeding in Patients With Atrial Fibrillation Using Diltiazem With Apixaban or Rivaroxaban

Wayne A. Ray, PhD; Cecilia P. Chung, MD, MPH; C. Michael Stein, MB, ChB; Walter Smalley, MD, MPH; Eli Zimmerman, MD; William D. Dupont, PhD; Adriana M. Hung, MD, MPH; James R. Daugherty, MS; Alyson Dickson, MA; Katherine T. Murray, MD

IMPORTANCE Diltiazem, a commonly prescribed ventricular rate-control medication for patients with atrial fibrillation, inhibits apixaban and rivaroxaban elimination, possibly causing overanticoagulation.

OBJECTIVE To compare serious bleeding risk for new users of apixaban or rivaroxaban with atrial fibrillation treated with diltiazem or metoprolol.

DESIGN, SETTING, AND PARTICIPANTS This retrospective cohort study included Medicare beneficiaries aged 65 years or older with atrial fibrillation who initiated apixaban or rivaroxaban use and also began treatment with diltiazem or metoprolol between January 1, 2012, and November 29, 2020. Patients were followed up to 365 days through November 30, 2020. Data were analyzed from August 2023 to February 2024.

EXPOSURES Diltiazem and metoprolol.

MAIN OUTCOMES AND MEASURES The primary outcome was a composite of bleeding-related hospitalization and death with recent evidence of bleeding. Secondary outcomes were ischemic stroke or systemic embolism, major ischemic or hemorrhagic events (ischemic stroke, systemic embolism, intracranial or fatal extracranial bleeding, or death with recent evidence of bleeding), and death without recent evidence of bleeding. Hazard ratios (HRs) and rate differences (RDs) were adjusted for covariate differences with overlap weighting.

RESULTS The study included 204 155 US Medicare beneficiaries, of whom 53 275 received diltiazem and 150 880 received metoprolol. Study patients (mean [SD] age, 76.9 [7.0] years; 52.7% female) had 90 927 person-years (PY) of follow-up (median, 120 [IQR, 59-281] days). Patients receiving diltiazem treatment had increased risk for the primary outcome (RD, 10.6 [95% CI, 7.0-14.2] per 1000 PY; HR, 1.21 [95% CI, 1.13-1.29]) and its components of bleeding-related hospitalization (RD, 8.2 [95% CI, 5.1-11.4] per 1000 PY; HR, 1.22 [95% CI, 1.13-1.31]) and death with recent evidence of bleeding (RD, 2.4 [95% CI, 0.6-4.2] per 1000 PY; HR, 1.19 [95% CI, 1.05-1.34]) compared with patients receiving metoprolol. Risk for the primary outcome with initial diltiazem doses exceeding 120 mg/d (RD, 15.1 [95% CI, 10.2-20.1] per 1000 PY; HR, 1.29 [95% CI, 1.19-1.39]) was greater than that for lower doses (RD, 6.7 [95% CI, 2.0-11.4] per 1000 PY; HR, 1.13 [95% CI, 1.04-1.24]). For doses exceeding 120 mg/d, the risk of major ischemic or hemorrhagic events was increased (HR, 1.14 [95% CI, 1.02-1.27]). Neither dose group had significant changes in the risk for ischemic stroke or systemic embolism or death without recent evidence of bleeding. When patients receiving high- and low-dose diltiazem treatment were directly compared, the HR for the primary outcome was 1.14 (95% CI, 1.02-1.26).

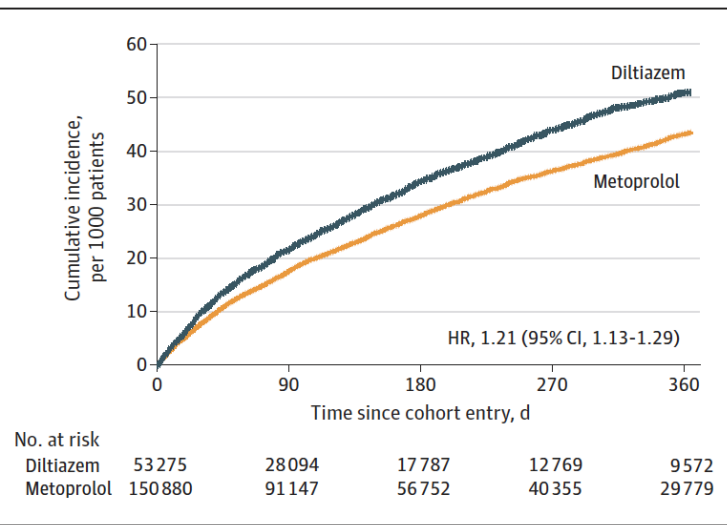
CONCLUSIONS AND RELEVANCE In Medicare patients with atrial fibrillation receiving apixaban or rivaroxaban, diltiazem was associated with greater risk of serious bleeding than metoprolol, particularly for diltiazem doses exceeding 120 mg/d.

[+ Supplemental content](#)

[+ CME Quiz at
jamacmelookup.com](#)

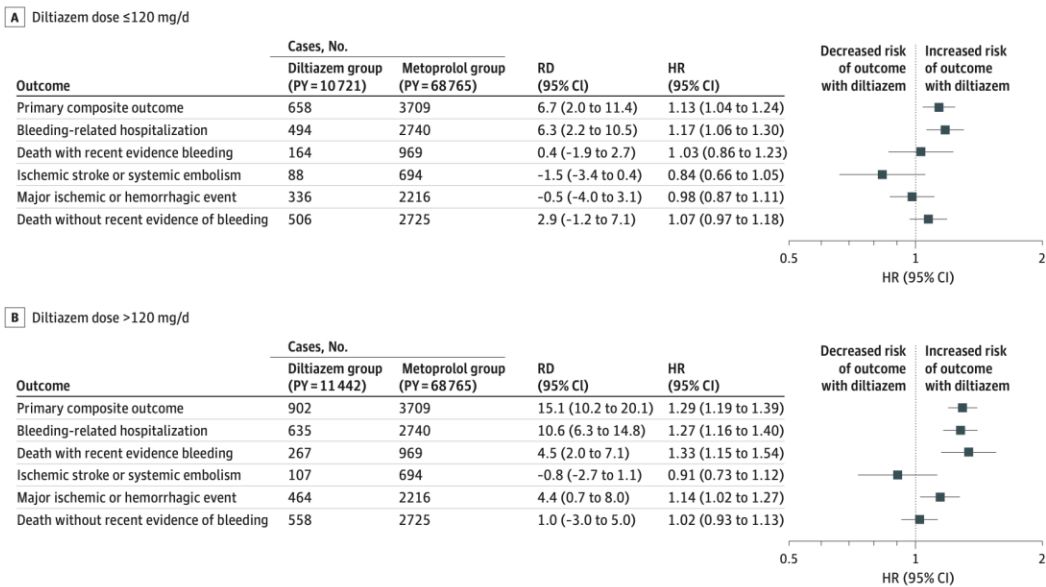
Ray AW et al.
JAMA 2024

Figure 1. Adjusted Cumulative Incidence of the Primary Composite Outcome According to Rate Control Treatment



The primary composite outcome was a bleeding-related hospitalization or death with recent evidence of bleeding. Analysis was adjusted for the covariates listed in eTable 7 in Supplement 1 with overlap weighting (eAppendix 5C in Supplement 1). Cumulative incidence was calculated as $1 - S(t)$, where S is the survival function estimated in a weighted proportional hazards analysis stratified by study ventricular rate-control drug and with no terms in the model. The median length of follow-up was 101 (IQR, 59-258) days for diltiazem and 120 (IQR, 59-290) days for metoprolol.

Figure 3. Adjusted Risk of Study Outcomes According to Initial Diltiazem Dose



Analysis was adjusted for the covariates listed in eTable 7 in Supplement 1 with overlap weighting. HR, indicates hazard ratio; PY, person-years; RD, rate difference.

Pharmaco-Genomics

- The cardiovascular drug-gene pair that has the most comprehensive pharmacokinetic, pharmacodynamic, observational, meta-analysis, and clinical trial data is clopidogrel-CYP2C19.
- This has led to a boxed warning in the drug labeling information for clopidogrel, an oral P2Y₁₂ inhibitor, that states that CYP2C19 poor metabolizers are at a higher risk for ischemic events when treated with clopidogrel
- These individuals can be identified by performing CYP2C19 genetic testing, and that they should be treated with an alternative therapy.
- Oral P2Y₁₂ inhibitor therapy among the most commonly used antiplatelet drugs, approved for use in patients with ACS, stroke, and peripheral arterial disease.

*Pereira NL et al.
Circulation 2024. AHA Scientific Statement.*

POC Test

- **1. VerifyNow-P2Y12 Assay:**
 - >180 - 376 PRU [P2Y12 Reaction Units] - No drug present
 - 10-180 PRU [P2Y12 Reaction Units] - Decreased platelet reactivity to P2Y12 inhibitor.
- **2. VerifyNow-Aspirin Assay:**
 - ≥ 550 Aspirin Reaction Units [ARU] - No evidence of platelet dysfunction due to Aspirin.
 - < 550 Aspirin Reaction Units [ARU] - Platelet dysfunction consistent with Aspirin - detected.

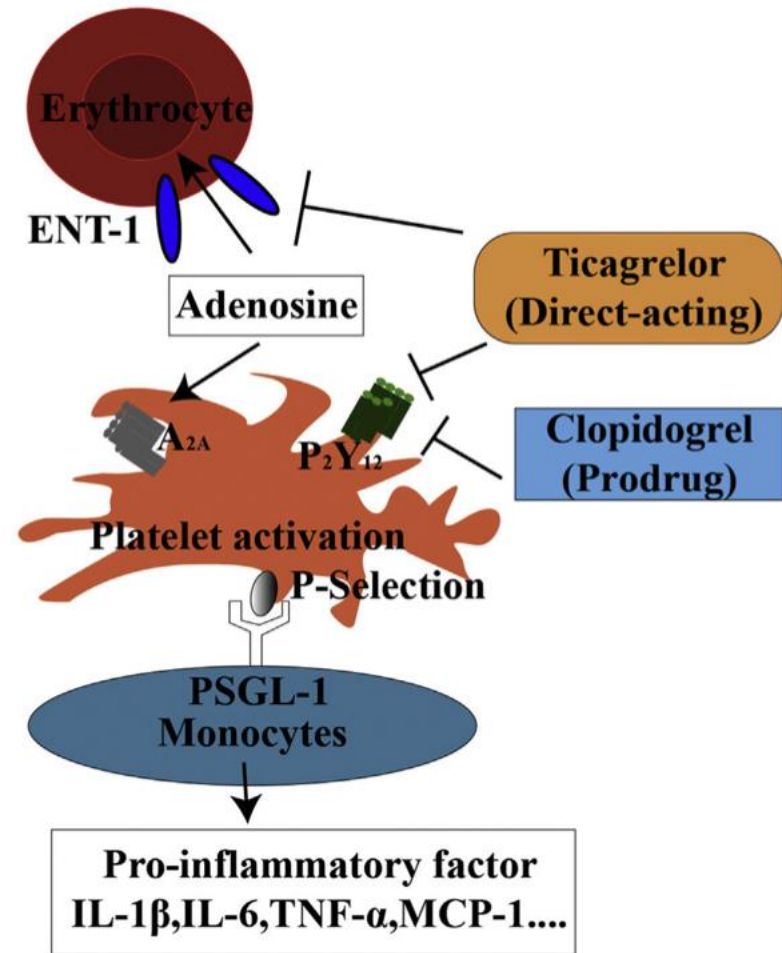


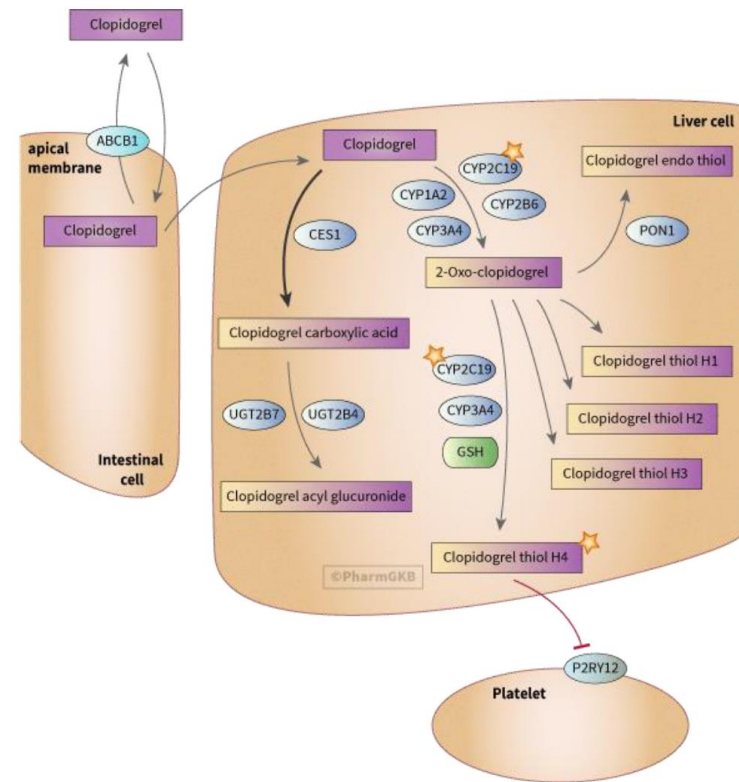
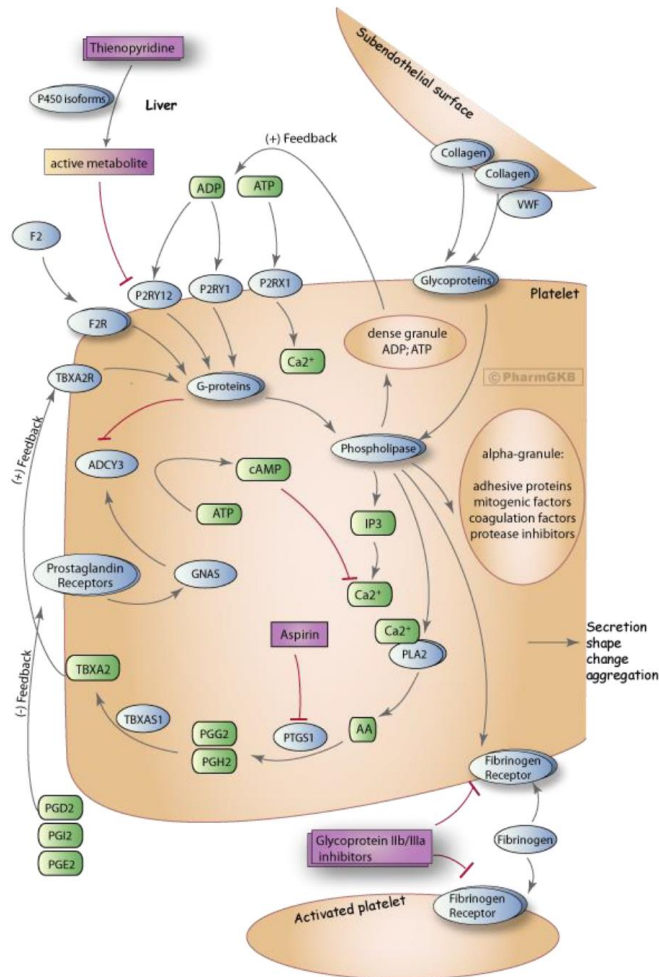
<https://practical-haemostasis.com/Platelets/VerifyNow%20Assay.html>

Differences in Pharmacological Mechanisms and Clinical Implications

Ticagrelor inhibits platelet activation and aggregation by reversibly interacting with the platelet P_2Y_{12} adenosine diphosphate (ADP) receptor to prevent signal transduction
<https://pmc.ncbi.nlm.nih.gov>

Direct-acting and does not require metabolic activation;
In contrast, clopidogrel is a pro-drug, 2-step metabolic activation in the liver; the active metabolite of clopidogrel binds covalently and irreversibly to the P_2Y_{12} receptor to block ADP binding.





Naveen L. Pereira. Circulation. CYP2C19 Genetic Testing for Oral P2Y12 Inhibitor Therapy: A Scientific Statement From the American Heart Association, Volume: 150, Issue: 6, Pages: e129-e150, DOI: (10.1161/CIR.0000000000001257)

Table 1. Genetic and Nongenetic Factors Affecting Pharmacokinetic Response to Oral P2Y12 Inhibitors			
	Clopidogrel	Prasugrel	Ticagrelor
Pharmacogenetics			
<i>ABCB1</i>	Decreased active metabolite levels in 3435T allele carriers ^{16,17} No association ^{18–21}	No association ¹⁸	...
<i>CYP1A2</i>	No association ^{21–23}	No association ^{23,24}	...
<i>CYP2B6</i>	Lower active metabolite levels in reduced function carriers ²² No association ²³	No association ^{23,24}	...
<i>CYP2C9</i>	Lower active metabolite levels in reduced function carriers ²⁴ No association ^{22,23}	No association ^{23,24}	...
<i>CYP2C19</i>	Active metabolite levels lower in reduced function carriers ^{17,18,22–31} Increased function variant (*17) not independently associated with active metabolite levels once *2 was accounted for ³²	No association ^{18,23–25,27,29,30}	...
<i>CYP3A4</i>	No association ^{21,23,33}	No association ^{23,24,33}	Ticagrelor and active metabolite levels associated with 2 SNPs (rs62471956, rs56324128) ³⁴ Active metabolite higher in *22 carriers ³³
<i>CYP3A5</i>	No association ^{19,22,23}	No association ^{23,24}	...
<i>PON1</i>	No association ^{19,22,23}	No association ¹⁸	...
<i>SLCO1B1</i>	...	...	Ticagrelor and active metabolite levels associated with SNP rs113681054, which is in linkage with functional variant rs4149056 ³⁴
<i>UGT2B7</i>	...	...	Active metabolite levels associated with rs61361928 ³⁴

Demographics			
Age	Minimal ^{19,35}	Higher active metabolite levels in patients >75 y of age ³⁶	Minimal ³⁷
BMI	Lower active metabolite levels in patients with higher body weight ³⁵	Higher active metabolite levels in patients with lower body weight ^{36,39}	Minimal ³⁷
Sex	Similar active metabolite levels between men and women ¹⁹	Similar between men and women ³⁹	Minimal ³⁷
Drug interactions			
Proton pump inhibitors	Lower clopidogrel active metabolite levels with omeprazole or esomeprazole; avoid concomitant use ^{40,41} Lansoprazole, dexlansoprazole, rabeprazole, and pantoprazole have minimal effect on the antiplatelet activity of clopidogrel. ^{40,42,43}	...	...
CYP3A inhibitors/inducers	Minimal interaction with statins ^{21,44}	Minimal ¹³	Strong CYP3A inhibitors (eg, ketoconazole, itraconazole) increased ticagrelor exposure; avoid concomitant use Strong CYP3A inducers (eg, rifampin, phenytoin) reduced ticagrelor exposure Ticagrelor increased exposure to simvastatin and lovastatin; avoid simvastatin/lovastatin doses >40 mg Ticagrelor FDA label ¹⁴
P-glycoprotein	...	...	Ticagrelor inhibits P-glycoprotein; monitor digoxin levels when initiating ⁴⁵
Comorbidities			
Diabetes	Lower active metabolite levels in patients with diabetes ⁴⁶	Lower active metabolite levels in patients with diabetes ⁴⁶	No impact ⁴⁷
Renal dysfunction	May inhibit conversion to active metabolite ¹⁷	Minimal impact ⁴⁸	No impact ¹⁷



Naveen L. Pereira. Circulation. CYP2C19 Genetic Testing for Oral P2Y12 Inhibitor Therapy: A Scientific Statement From the American Heart Association, Volume: 150, Issue: 6, Pages: e129-e150, DOI: (10.1161/CIR.0000000000001257)

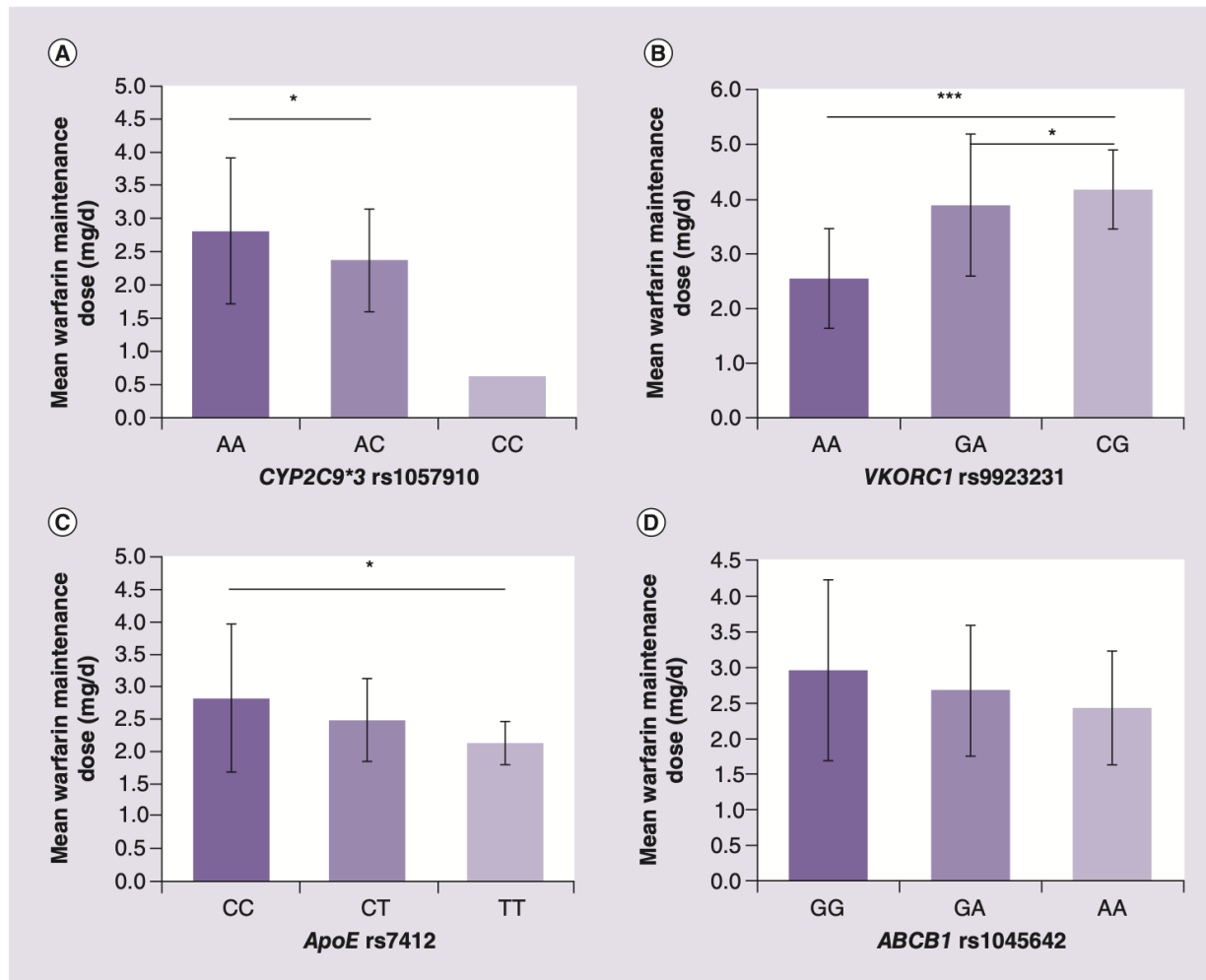


Figure 1. Mean stable warfarin dose by genotypes in elderly Shanghai Han Chinese patients (mg/d). (A) *CYP2C93 rs1057910 genotype. (B) *VKORC1* rs9923231 genotype. (C) *ApoE* rs7412 genotype. (D) *ABCB1* rs1045642 genotype. * $p < 0.05$; *** $p < 0.001$ (analyzed by one-way analysis of variance).**

Time in Therapeutic Range (TTR) in Warfarinization

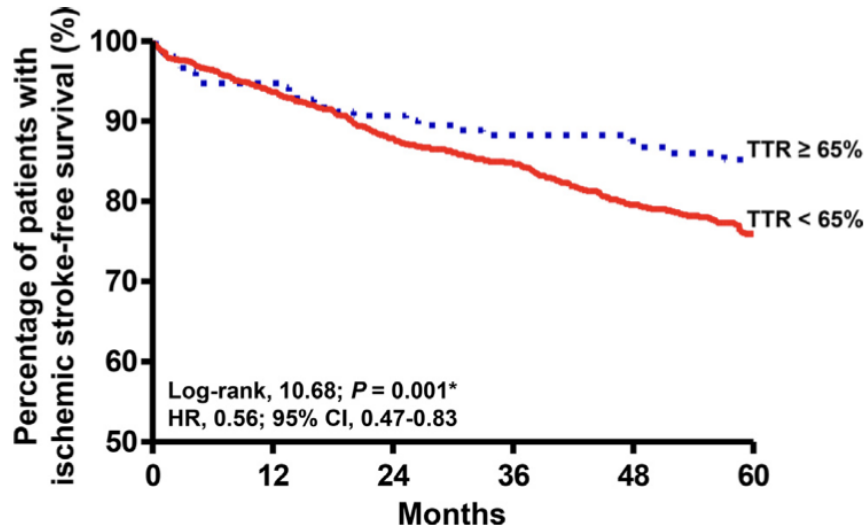


Figure 1. Kaplan-Meier estimates of ischemic stroke-free survival patients with atrial fibrillation with time in therapeutic range (TTR) $\geq$ 65% and TTR < 65%. CI, confidence interval; HR, hazard ratio. * $P < 0.05$.

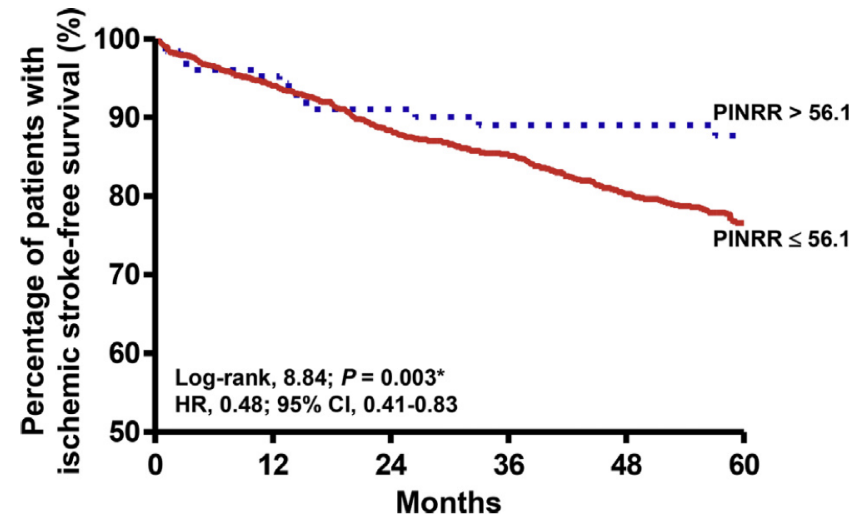


Figure 3. Kaplan-Meier estimates of ischemic stroke-free survival in patients with atrial fibrillation with percentage of international normalized ratio in therapeutic range (PINRR) > 56.1% and PINRR ≤ 56.1%. CI, confidence interval; HR, hazard ratio. * $P < 0.05$.

Conclusions

- Antithrombotic therapies, including antiplatelet and anticoagulant drugs have clinically proven efficacies in the protection against CVD
- Understanding the pharmacological principles of each therapy allows clinicians to decide the optimal therapeutic strategies according to specific patient scenario
- International clinical guidelines are available for a majority of these clinical scenario; individualized assessments are paramount
- Patient factors and comorbidities need to be taken into consideration
- Drug interactions could result in clinically significant consequences to therapies and a sensible degree of caution is necessary
- Evidence rapidly evolving and continuous professional updates are necessary

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